



LABORATORY MANUAL

SA 101 SYSTEM ADMINISTRATION AND MAINTENANCE





A LABORATORY MANUAL IN SA 101 SYSTEM ADMINISTRATION AND MAINTENANCE

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PREFACE

This laboratory manual for System Administration and Maintenance was created to make learning more practical and easier to connect with. It walks students through simple, hands-on activities that help them develop the skills needed to manage and maintain computer systems. With technology constantly evolving, system administrators play a crucial role in maintaining the security, reliability, and optimal functioning of computers and networks for the users who rely on them. This manual aims to support students in that journey by providing precise and practical exercises they can follow with confidence.

Each activity is designed to bring classroom lessons to life. Instead of learning purely through discussions, students will have the opportunity to install systems, adjust settings, manage user accounts, troubleshoot problems, and perform other essential maintenance tasks independently. These activities help them understand not only *how* things work, but also *why* these tasks are crucial to real IT environments.

More importantly, this manual aims to build the students' confidence. System administration can be challenging, especially at the beginning. Still, with patience and hands-on practice, students will learn to solve problems, make sound decisions, and handle responsibilities similar to those they will face in actual workplaces. As they complete each exercise, they will also develop stronger critical thinking skills, improve their time management, and learn to approach technical tasks with patience, professionalism, and a calm mindset.

Ultimately, this manual serves as a dependable companion throughout the course—one that guides and supports students as they develop into capable future IT professionals. Every lab activity encourages them to explore, experiment, and strengthen the essential skills they will carry with them into their careers.



LABORATORY SAFETY PROCEDURES

To keep our lab safe, organized, and comfortable for everyone, please follow these simple guidelines while working on your system administration tasks.

A. General Safety

1. Enter and leave the lab quietly and respectfully.
2. No food or drinks near the computers and equipment.
3. Handle all devices gently—treat them as if they were your own.
4. Don't open or repair any hardware unless your instructor gives permission.
5. Keep your workstation clean to work smoothly and prevent accidents.

B. Electrical and Equipment Safety

1. Make sure your hands are dry before touching any cables or equipment.
2. Avoid unplugging or rearranging wires unless instructed to do so by your instructor.
3. Always shut down computers properly before turning them off or unplugging them from the power source.
4. Report anything unusual—such as loose cables, strange odors, or overheating devices.
5. Don't remove covers or touch internal parts of computers or servers without supervision.

C. System, Data, and Software Safety

1. Only use the software and tools approved for the activity.
2. Don't change settings or configurations unless the lab instructions require it.
3. Back up your files and work regularly to avoid losing progress.
4. Keep your passwords and login details private.
5. Avoid installing unknown apps, opening suspicious files, or running scripts you are unsure of.

**D. Behavior, Professionalism, and Responsible Use**

1. Respect your classmates—never access their accounts or files.
2. Don't try to get around restrictions, access areas you're not supposed to, or use privileges that weren't assigned to you.
3. Be mindful of your actions—avoid anything that could accidentally interrupt or damage the system.
4. Be honest with your work. Don't copy others' outputs or misuse administrative tools in ways they weren't meant to be used.
5. And most importantly, follow the school's IT guidelines and handle all systems responsibly and ethically.



LABORATORY SAFETY AGREEMENT

I, the undersigned, have read and fully understood the **Laboratory Safety Procedures** for the System Administration and Maintenance course. I agree to follow all the safety guidelines, rules, and responsibilities described in the document.

I understand that these procedures are designed to maintain a safe, organized, and productive laboratory environment for everyone. I acknowledge that failing to follow any of the safety rules may lead to disciplinary action, loss of lab privileges, or other academic consequences.

By signing below, I am committing to being responsible, respectful, and safety-conscious throughout the course.

Name (Printed): _____

Course/Year/Section: _____

Signature: _____

Date: _____

Instructor's Name: Engr. Raymond Bryan M. Esmenda

Department: Information Technology Department

Institution: Pangasinan State University – Lingayen Campus



RUBRIC FOR CHECKING LABORATORY ACTIVITY

Criteria	Description	Points
1. Completion (30%)	The laboratory activity is fully completed according to the given instructions, procedures, and learning objectives. All required outputs are submitted.	30
2. Accuracy and Functionality (25%)	Outputs are technically accurate, error-free, and function as required. Configurations, commands, and procedures are correct and verified.	25
3. Code Quality and Organization (20%)	Steps are clearly and logically followed. Screenshots are correctly labeled. Settings and configurations follow best practices and proper formatting.	20
4. Understanding and Application (15%)	Student demonstrates a clear understanding of the concepts and can explain how the configuration works. Reflections or explanations show correct application of theory.	15
5. Punctuality and Submission (10%)	Work is submitted on time, complete, and formatted adequately following the laboratory manual template.	10
TOTAL		100

Grading Scale:

- 96–100: Excellent
- 91–95: Very Good
- 86–90: Good
- 80–85: Satisfactory
- Below 80: Needs Improvement



INTRODUCTION

Computer systems are now part of almost everything we do—from school and work to business and communication. Because of this, it has become essential to have people who know how to keep these systems running smoothly and safely. This is what System Administration and Maintenance is all about. It teaches students how to install, manage, and maintain the computers and networks that organizations depend on every day.

At Pangasinan State University (PSU), the BSIT program aims to equip students with the skills necessary to handle real-world technical tasks. SAM helps build those skills by introducing learners to the day-to-day responsibilities of a system administrator, such as managing user accounts, organizing files, resolving errors, updating systems, and ensuring everything works as expected.

This laboratory manual was created to guide students through activities that let them learn by doing. Instead of just reading or listening, they will actually work on systems—setting them up, customizing them, and solving common problems. These exercises reflect real situations they might face in the workplace and are designed to be clear, practical, and easy to follow.

The primary objective of this manual is to help students develop confidence as they learn how systems operate and how to maintain them effectively. By participating in every activity, they will develop the habits, skills, and mindset necessary to support reliable and secure computer environments in the future.



PURPOSE OF THE MANUAL

This manual was created to help students learn System Administration and Maintenance in a practical and hands-on way. Instead of just reading about concepts in class, this manual enables students to work with systems, try out various approaches, and understand how real computer environments are managed. Its primary purposes are to:

1. Guide students through clear, step-by-step activities such as installing systems, managing users, troubleshooting problems, performing backups, and setting up basic network configurations.
2. Strengthen their understanding of the essential tasks and responsibilities of a system administrator through practical exercises.
3. Help students learn how to identify and resolve common system issues, keep computers running smoothly, and maintain a safe and reliable environment.
4. Encourage them to think independently, work well with others, make responsible decisions, and solve problems based on real situations.
5. Give instructors a valuable reference for checking student progress, evaluating skills, and assessing performance during laboratory activities.

Overall, this manual aims to make learning more engaging and meaningful by allowing students to apply what they learn directly to real-world systems. Through hands-on practice, they will build the confidence and skills needed to support stable and secure IT operations in the future.



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SYSTEM ADMINISTRATION AND MAINTENANCE

INTRODUCTION TO SYSTEM ADMINISTRATION

Q INTRODUCTION

System administration is all about ensuring that computer systems and networks operate as they should. It includes setting them up, keeping everything up to date, and resolving any problems that arise. System administrators play a crucial role in maintaining system security and ensuring that all components function smoothly. Because of their work, people in an organization can focus on their tasks without interruptions. In this activity, you are introduced to the core concepts of system administration—what administrators do, the components that make up an IT setup, and the policies that guide the management of technology within an organization.

Q SCOPE

This laboratory activity focuses on the basic understanding of system administrator duties, network infrastructure components, and IT policies. Students will explore the fundamental responsibilities and services involved in managing an organization's information systems.

Q MATERIALS

- Laptop or desktop computer
- Internet connection
- Word processing application (MS Word/Google Docs)
- Access to online references
- Optional: Diagramming tool (Draw.io, Lucidchart, Canva)

Q LEARNING OUTCOMES

At the end of this activity, the students should be able to:

1. Define system administration and its role in IT environments.
2. Identify the significant duties of a system administrator.
3. Explain the function of essential network and infrastructure services.
4. Analyze the importance of organizational IT policies.
5. Demonstrate understanding through tasks and guided questions.

LABORATORY ACTIVITY 1: INTRODUCTION TO SYSTEM ADMINISTRATION

Objectives:

At the end of this activity, the students should be able to:

1. Understand what system administration is and its role within an organization.
2. Identify common server types—like file servers, domain controllers, and application servers—and know what they're used for.
3. Recognize essential network services, such as DNS, DHCP, and Active Directory, and understand their importance.
4. Sketch a simple IT infrastructure showing how servers, devices, and network equipment work together.
5. Learn the core responsibilities and skills needed to manage and maintain enterprise-level systems.

BASIC INFORMATION

System administration ensures the reliability, continuity, and security of IT systems within an organization. This section provides a structured overview of key concepts needed to understand the system administrator's role.

Overview of System Administration

System administration includes managing software, hardware, user accounts, security controls, and troubleshooting to maintain operational stability.

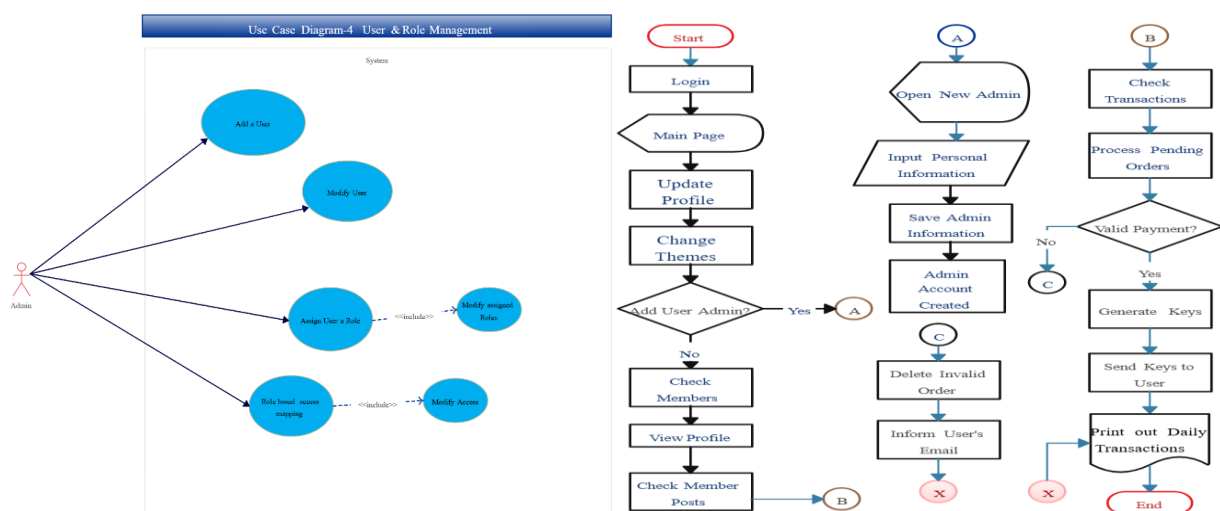


Figure 1. Vector-style overview of system administration responsibilities.

Key Duties of a System Administrator

SYSTEM ADMINISTRATOR

Duties of a System Administrator

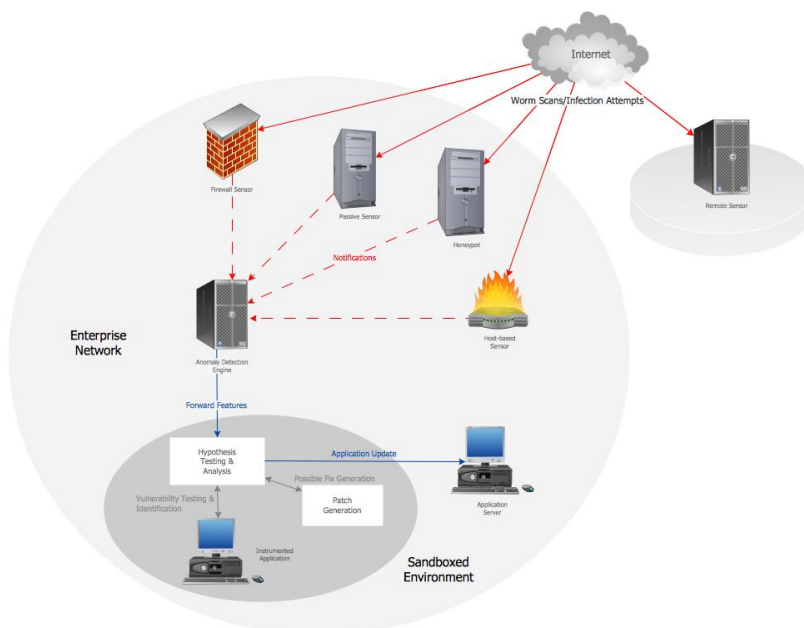


System Administrator

Figure 2. Visual representation of common system administrator duties.

The core duties of system administration include managing user accounts and permissions, performing system updates and patching, monitoring servers and networks, configuring security measures and mitigating threats, planning for backups and recovery, and maintaining documentation while enforcing organizational policies.

Network and Infrastructure Services



Essential services include:

- DNS
- DHCP
- File Sharing
- Directory Services
- Email Servers
- Authentication Systems

Figure 3. Network and infrastructure services diagram.

Organizational IT Policies and Standards

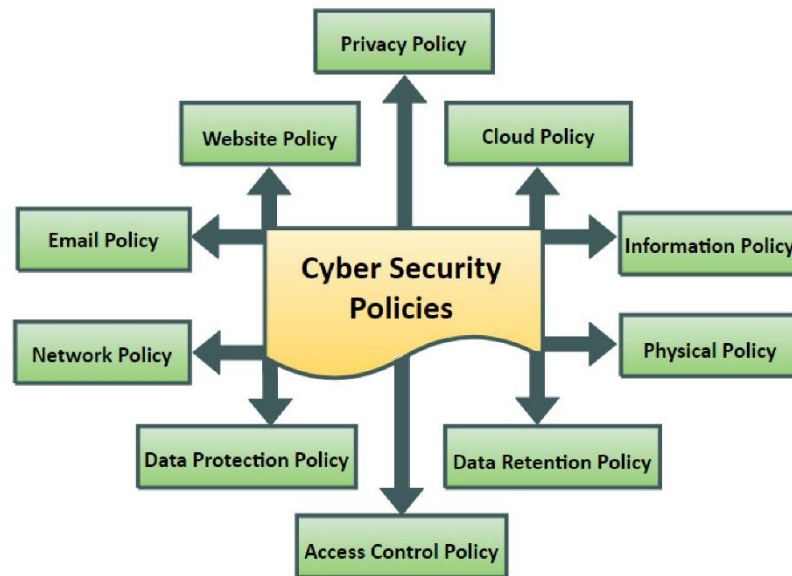


Figure 4. Sample illustration of IT policy documentation.

Policies ensure consistent, secure, and responsible use of organizational technology. System administrators enforce these to maintain compliance and protect data.

Importance of System Administration in Business Operations

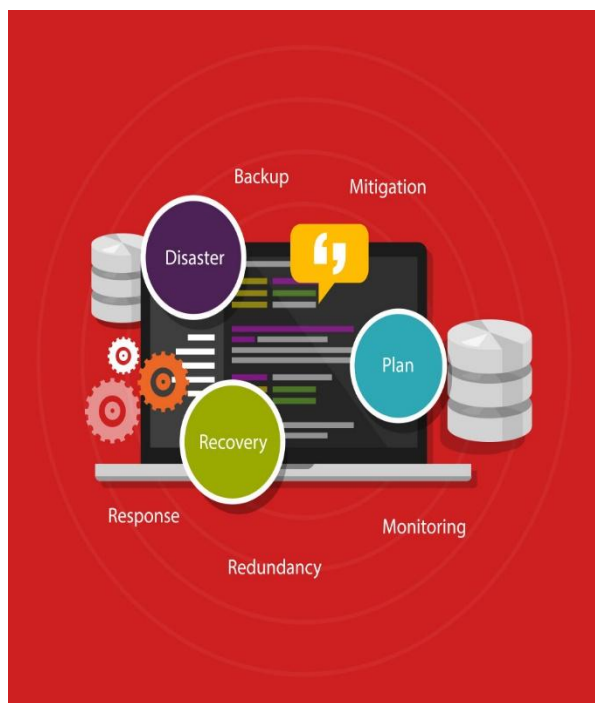


Figure 5. Diagram showing how system administration supports business continuity.

Effective system administration ensures:

- Minimal downtime
- Security and data protection
- Smooth and reliable operations
- Compliance with regulations
- Continuous productivity

Task #1: IDENTIFYING SYSTEM ADMINISTRATOR DUTIES

Instructions:

1. Research and list at least **eight (8)** system administrator responsibilities.
2. Categorize each under:
 - User Management
 - System Maintenance
 - Network & Infrastructure
 - Security Administration
3. Provide short descriptions and examples.

Output Example:

Category	Responsibility	Description / Example
User Management	Account creation	Adding new employees to Active Directory

TASK #2: NETWORK AND INFRASTRUCTURE SERVICES ANALYSIS

Instructions:

1. Identify **five (5)** infrastructure services managed by system administrators.
2. Explain the purpose and importance of each.

Image for Task Reference

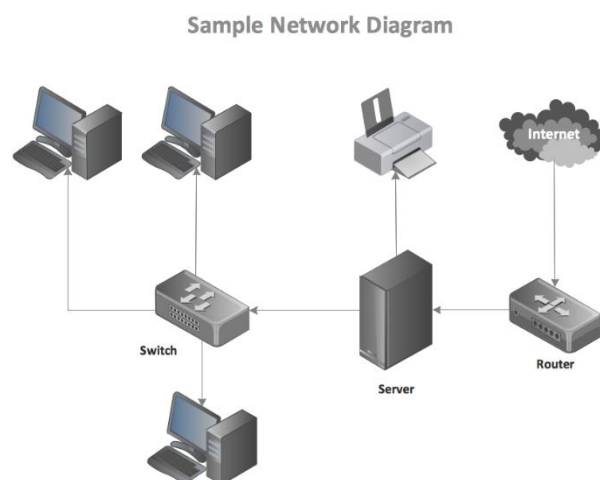


Figure 6. Basic illustration of server-client IT infrastructure.



TASK #3: REVIEW OF ORGANIZATIONAL IT POLICIES

Instructions:

1. Review sample IT policies such as Acceptable Use and Password Policy.
2. Write a **150–200-word** summary explaining:
 - Why IT policies are important?
 - How they protect organizations?
 - How system administrators enforce them?



GUIDED QUESTIONS

1. Why is system administration essential in organizations?
2. Which system administrator duty is the most critical and why?
3. How do network services enable productivity?
4. Why must IT policies be strictly enforced?
5. How does Chapter 1 prepare you for server installation (Chapter 2)?



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Cisco Systems. (2020). Introduction to networking: A beginner's guide. Cisco Press.

Silberschatz, A., Gagne, G., & Galvin, P. (2018). Operating system concepts (10th ed.). Wiley.

Turban, E., Pollard, C., & Wood, G. (2018). Information technology for management (11th ed.). Wiley.



SYSTEM ADMINISTRATION AND MAINTENANCE

INSTALLING THE VIRTUAL SERVER ENVIRONMENT (VIRTUALBOX SETUP)

Q INTRODUCTION

Virtualization enables system administrators to run and manage multiple operating systems on a single physical computer. Using Oracle VM VirtualBox, students can create secure and convenient virtual environments that mimic real servers. In this activity, you'll learn how to install VirtualBox, set up your own virtual machines, and confirm that both Windows 10 and Windows Server are installed and working correctly.

Q SCOPE

This laboratory activity includes:

- Installing Oracle VM VirtualBox
- Navigating the VirtualBox interface
- Creating a Windows 10 VM
- Verifying that Windows 10 runs properly
- Installing Windows Server ISO with GUI

Q MATERIALS

- Laptop / Desktop Computer
- Stable Internet Connection
- Oracle VM VirtualBox Installer
- Windows 10 ISO (for testing)
- Windows Server ISO (for GUI installation)
- 8 GB RAM minimum
- 30–50 GB available disk space

Q LEARNING OUTCOMES

At the end of this activity, the students should be able to:

1. Install Oracle VM VirtualBox properly.



2. Navigate and recognize components of the VirtualBox interface.
3. Create and configure a Windows 10 virtual machine.
4. Verify that Windows 10 runs inside a VM.
5. Install Windows Server with Desktop Experience (GUI).

LABORATORY ACTIVITY 2: INSTALLING THE VIRTUAL SERVER ENVIRONMENT (VIRTUALBOX SETUP)

Objectives:

At the end of this activity, the students should be able to:

1. Understand what virtualization is and why it's essential in modern system administration and server setup.
2. Install Oracle VM VirtualBox and get comfortable with its interface, tools, and basic settings.
3. Create a working virtual machine by choosing the correct memory, storage, and system settings for the operating system you'll use.
4. Install a guest operating system, such as Windows 10, and ensure it runs properly within VirtualBox.
5. Demonstrate your ability to set up and manage a Windows Server environment, including mounting ISO files and preparing the server for future configuration tasks.

BASIC INFORMATION

Virtualization enables system administrators to test, deploy, and manage multiple server environments without the need for separate physical machines. This section introduces the concepts and components required to set up a virtual environment.

What is Virtualization?

Virtualization makes it possible to run several operating systems on a single physical computer by creating virtual hardware for each one.

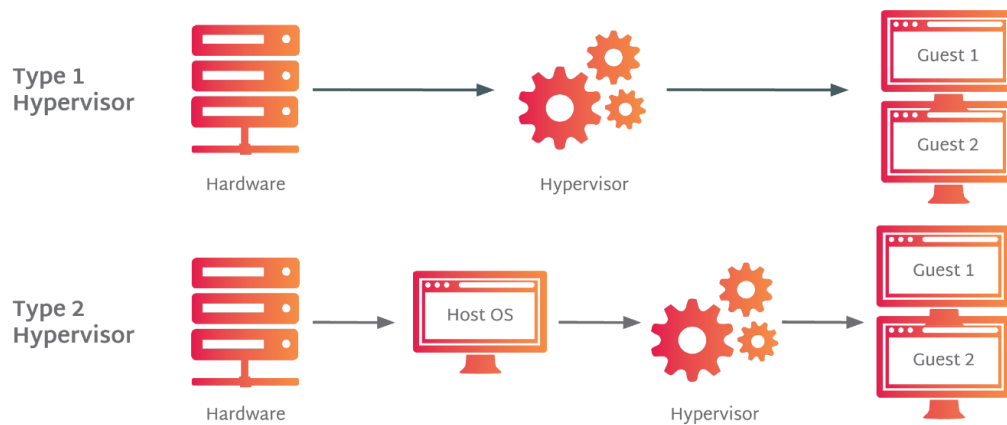


Figure 7. The virtualization concept shows a host machine running multiple virtual machines.

Introduction to Oracle VM VirtualBox

VirtualBox is an open-source Type-2 hypervisor that allows users to create and manage virtual machines.



Figure 8. Illustration of the Oracle VM VirtualBox.

Components of a Virtual Machine

A VM simulates hardware components such as CPUs, RAM, storage, displays, and network adapters, all of which are configurable by the user.

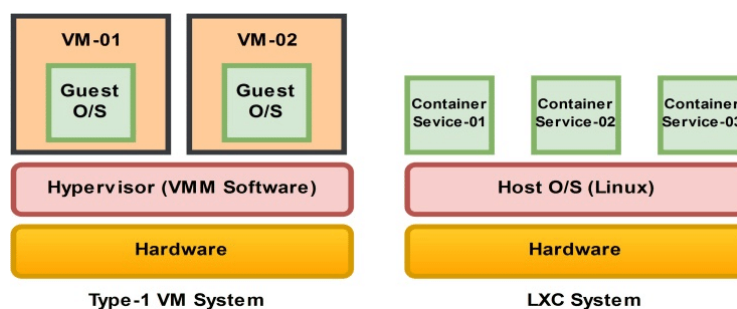


Figure 9. Virtual machine components inside VirtualBox.

Benefits of Virtualization

Virtual machines reduce hardware cost, improve system testing, allow safe experimentation, and provide quick system recovery through snapshots.



Figure 10. Benefits of virtualization for system administration.

TASK #1: EXPLORING THE VIRTUALBOX INTERFACE

Instructions:

Students download and install VirtualBox on their computer.



Figure 11. Visual example of the VirtualBox installation wizard.

TASK #2: INSTALLING ORACLE VM VIRTUALBOX

Instructions:

Students explore the toolbar, VM list, settings panel, and global tools.

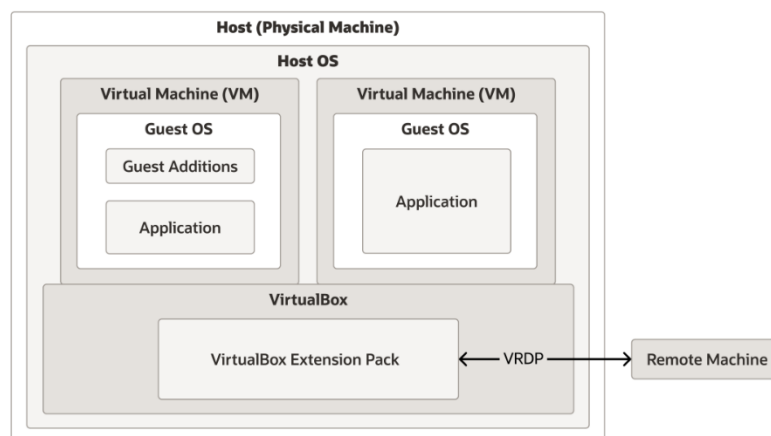


Figure 12. Diagram of the VirtualBox interface.

TASK #3: CREATING YOUR FIRST WINDOWS 10 VIRTUAL MACHINE

Instructions:

1. Start VM Creation

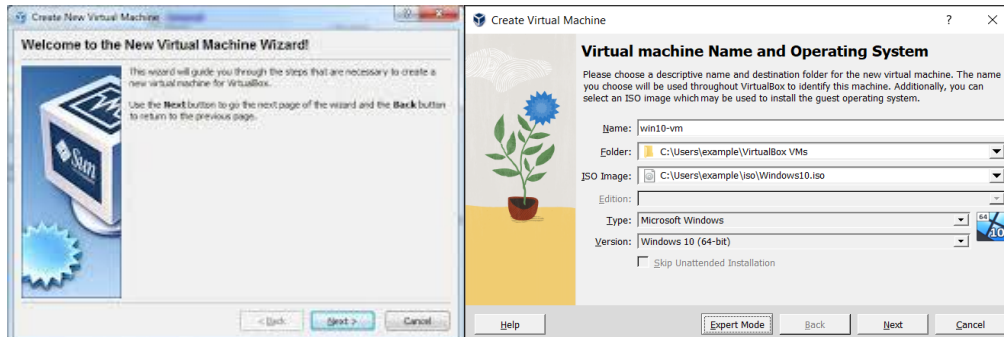


Figure 13. The "Create Virtual Machine" wizard.

2. Enter Windows 10 Details

Fill out required fields:

- **Name:** Windows10-VM
- **Type:** Microsoft Windows
- **Version:** Windows 10 (64-bit)

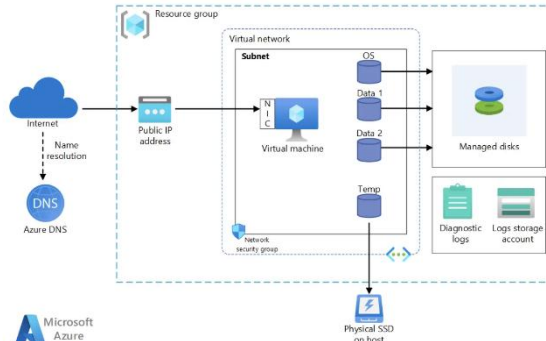
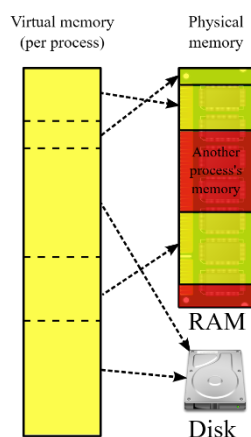


Figure 14. Selecting Windows 10 (64-bit) in VM settings.

3. Allocate Virtual Hardware



Recommended:

- **Memory:** 4096 MB
- **Processors:** 1–2 CPUs

Figure 15. Allocating RAM and CPU to the VM

4. Create Virtual Hard Disk

- **Type:** VDI
- **Storage:** Dynamically allocated
- **Size:** 30 GB minimum

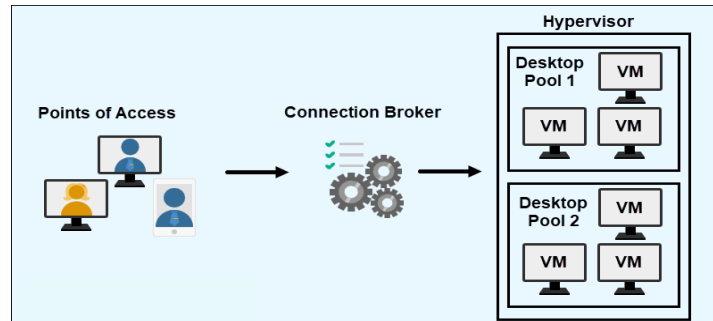


Figure 16. Creating a virtual hard disk.

TASK #4: Verifying Windows 10 Boots & Runs

Instructions:

1. Start the Windows10-VM

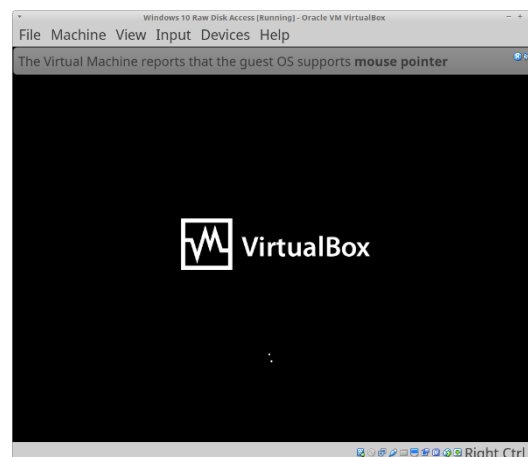


Figure 17. Windows 10 booting inside VirtualBox.

2. Confirm Windows Desktop Loads

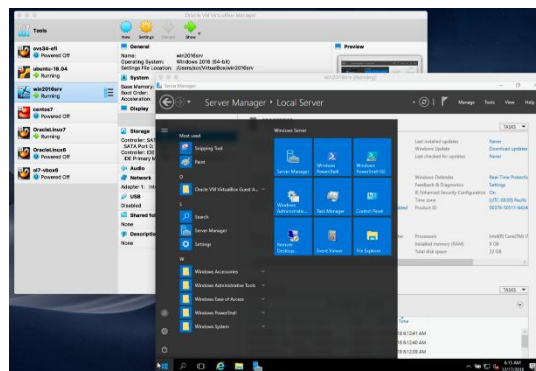


Figure 18. Windows 10 desktop successfully running inside VirtualBox.

Students must include these screenshots in their documentation.

TASK #5: Installing Windows Server ISO (WITH GUI)

Instructions:

1. Mount the Windows Server ISO

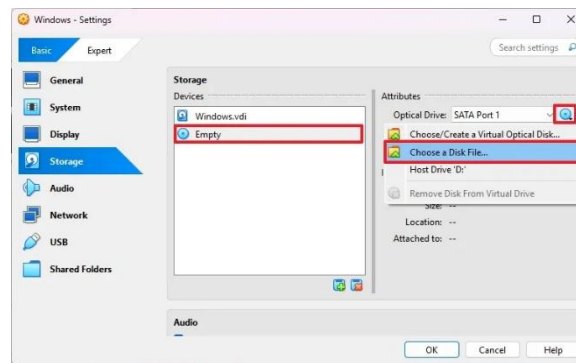


Figure 19. Mounting Windows Server ISO.

2. Start Installation

Select:

- **Language:** English
- **Keyboard:** US
- **Click:** Install Now



Figure 20. Windows Server initial setup.

3. Choose GUI Version

Select:

- ✓ **Windows Server Standard (Desktop Experience)**
- ✓ **or Windows Server Datacenter (Desktop Experience)**

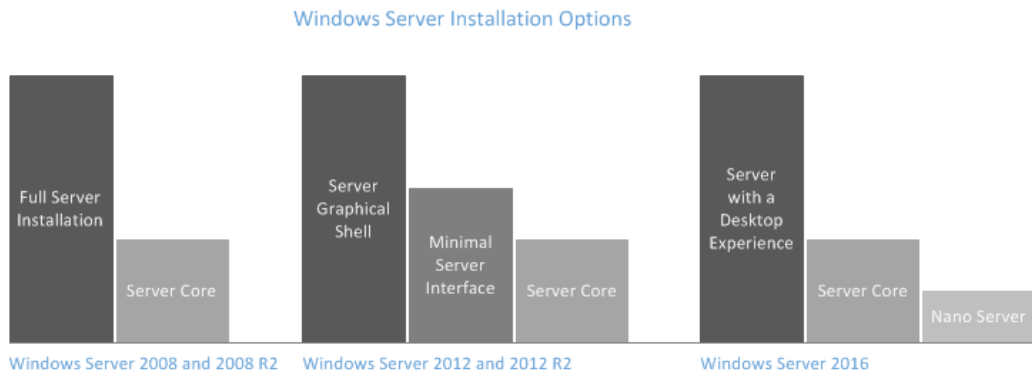


Figure 21. Choosing Windows Server with GUI.

4. Partition and Install

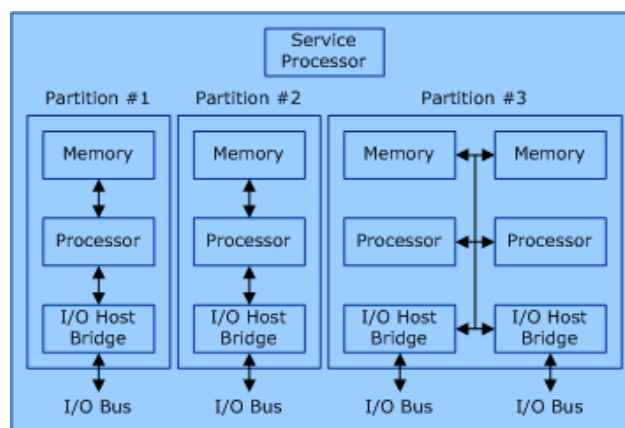


Figure 22. Selecting disk partition for installation.

5. Confirm GUI Desktop Loads

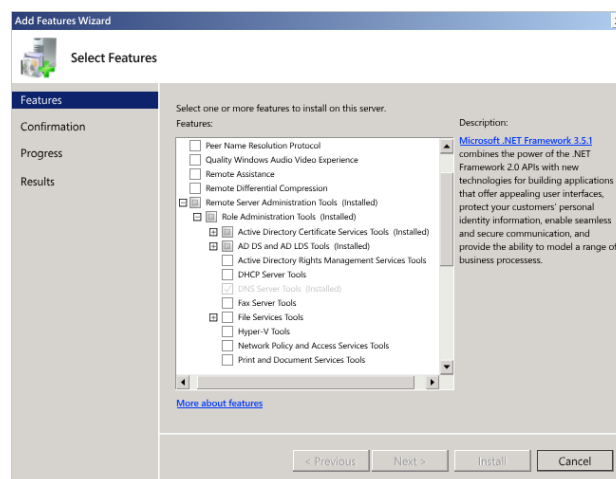


Figure 23. Windows Server GUI environment after installation.



GUIDED QUESTIONS

1. What's the difference between "Desktop Experience" and "Server Core"?
2. Why do some organizations still prefer using servers with a graphical interface (GUI)?
3. What is the purpose of mounting an ISO file when working in a virtual machine?
4. How is managing Windows Server different from managing Windows 10?
5. Why does the boot order of a virtual machine matter during installation?



REFERENCES

Oracle. (2023). Oracle VM VirtualBox User Manual. Oracle Documentation.
<https://www.virtualbox.org/manual/>

Microsoft. (2023). Windows troubleshooting tools. Microsoft Learn.
<https://learn.microsoft.com/en-us/windows-server/administration/windows-commands/>

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SYSTEM ADMINISTRATION AND MAINTENANCE

WINDOWS SERVER CONFIGURATION & SERVER ROLES



INTRODUCTION

Windows Server serves as the backbone for managing centralized authentication, resource sharing, network services, and other enterprise-level operations. Once Windows Server is installed with a GUI, the next step is to prepare it for actual organizational use. This includes tasks such as renaming the server, setting up network settings, enabling remote management, and installing essential roles, including Active Directory Domain Services (AD DS), DHCP, and DNS.

In this laboratory activity, students will explore the Windows Server Manager interface and learn how to perform basic post-installation setup and configure essential server roles.



SCOPE

This activity covers:

- Initial post-installation configuration of Windows Server
- Server renaming and system identification
- Configuring network settings
- Enabling Remote Desktop
- Introduction to Server Manager
- Installing and verifying key server roles



MATERIALS

- Installed Windows Server VM (Desktop Experience)
- Stable Internet connection
- Administrator account access
- Word processing application (for documentation)



LEARNING OUTCOMES

At the end of this activity, students will be able to:

1. Perform initial configuration of Windows Server.
2. Modify system identity and networking.
3. Navigate and use **Server Manager** effectively.
4. Install and configure fundamental server roles.
5. Document server configuration tasks with screenshots.

LABORATORY ACTIVITY 3: WINDOWS SERVER CONFIGURATION & SERVER ROLES

Objectives:

At the end of this activity, the students should be able to:

1. Set up Windows Server for the first time by giving it an appropriate name and configuring the key settings it needs to run smoothly.
2. Update the server's identity and network settings, including assigning a static IP address and enabling remote access.
3. Explore and use Server Manager to view system information, manage server roles, and access essential administrative tools.
4. Install and configure important server roles such as Active Directory Domain Services (AD DS), DNS, and DHCP.
5. Document your work by taking clear screenshots of each configuration step to show exactly what you did and the results of your changes.

BASIC INFORMATION

This section explains the key concepts needed before configuring Windows Server.

What is Server Configuration?

Server configuration refers to the process of preparing a newly installed server for operational use. This includes setting the server's name, network identity, and updates, as well as preparing the environment for role deployment.

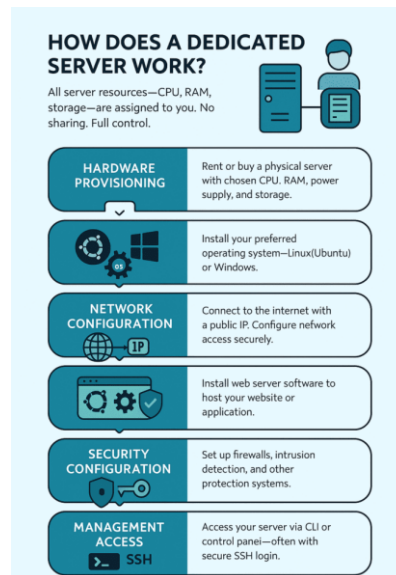


Figure 24. Overview of server configuration steps.

Introduction to Server Manager

Server Manager is the central management tool in Windows Server. It is used for adding roles, managing features, viewing system health, and performing administrative tasks.

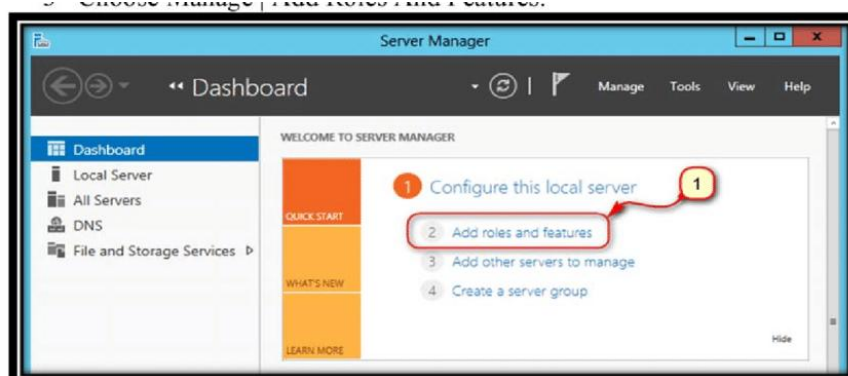


Figure 25. Server Manager dashboard overview.

Server Roles in Windows Server

Server roles are specialized functionalities that a server can perform, such as:

- Active Directory Domain Services (AD DS)
- DNS Server
- DHCP Server
- File and Storage Services
- Remote Desktop Services

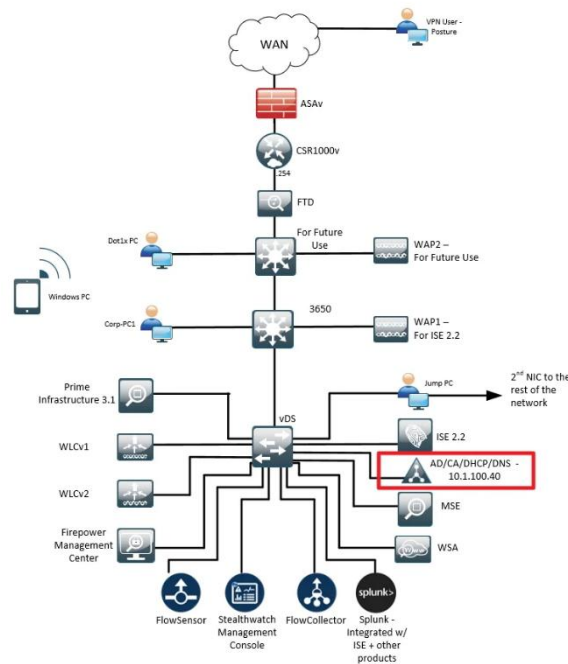


Figure 26. Common server roles in Windows Server.

Why Server Roles Matter?

Server roles enable centralized authentication, IP management, resource sharing, and enterprise-grade network services. Proper configuration is essential for secure and efficient IT environments.

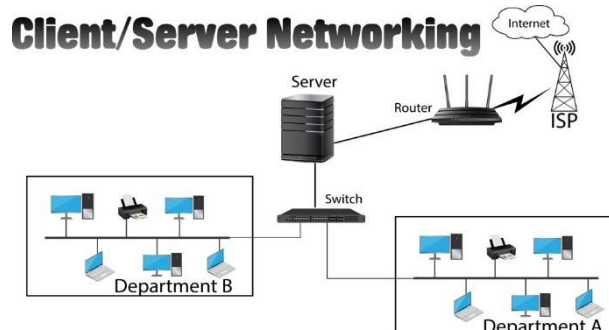


Figure 27. Server roles supporting network and enterprise operations.

TASK #1: PERFORMING INITIAL SERVER CONFIGURATION

Instructions:

1. Rename the Server

- Open Server Manager
- Click **Local Server**
- Select **Computer name** → Change
- Rename to:
SRV-YourLastName (example: SRV-DelaCruz)
- Restart the server.

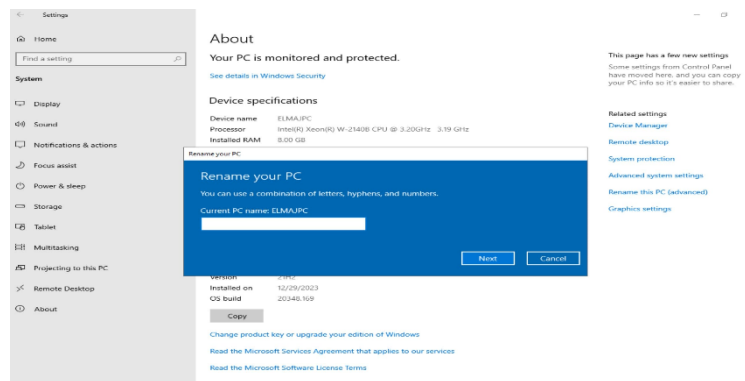


Figure 28. Renaming the Windows Server machine.

2. Configure Network Settings

- Go to **Network & Internet Settings**
- Set a **static IP address**
- Assign:
 - IP Address: 192.168.X.X
 - Subnet Mask: 255.255.255.0
 - Default Gateway: your router address
 - Preferred DNS: 127.0.0.1 (local)

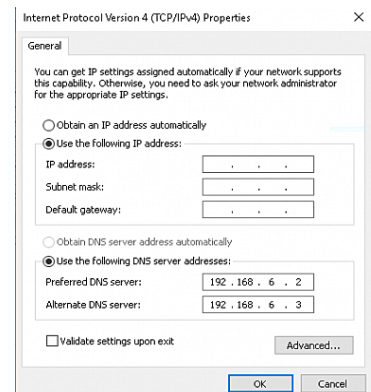


Figure 29. Setting static IPv4 network configuration.

3. Enable Remote Desktop

- Open **Server Manager** → **Local Server**
- Set **Remote Desktop** → **Enabled**
- Allow connections from **Network Level Authentication (NLA)**

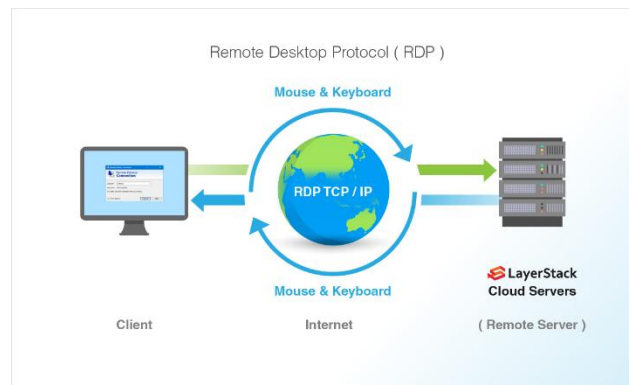


Figure 30. Enabling Remote Desktop access.

TASK #2: INTRODUCTION TO SERVER MANAGER

Instructions:

1. Open **Server Manager**
2. Identify the following sections:

- Dashboard
- Local Server
- All Servers
- Tools

3. Write a **short description** (2–3 sentences) of each section.

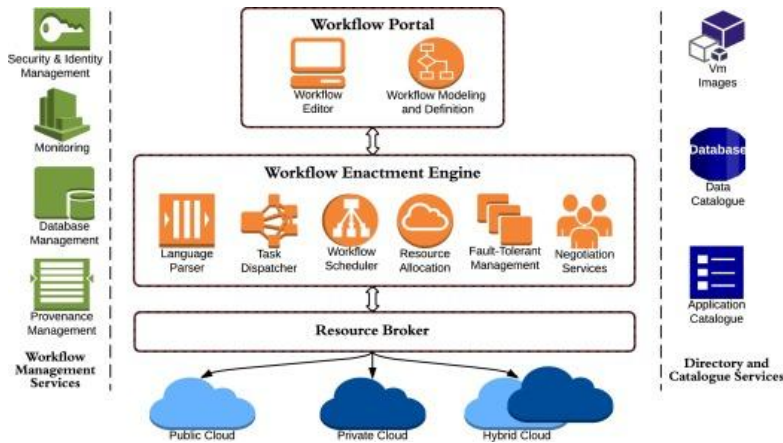


Figure 31. Server Manager interface sections.

ASK #3: INSTALLING WINDOWS SERVER ROLES

Instructions:

Students will install and document the following roles:

- ✓ Active Directory Domain Services (AD DS)
- ✓ DNS Server
- ✓ DHCP Server

1. Add Roles and Features

- Open **Server Manager**
- Click **Manage** → **Add Roles and Features**
- Choose **Role-based or feature-based installation**
- **Select your local server**

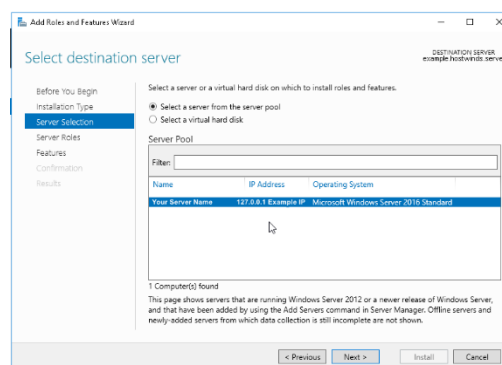


Figure 32. Add Roles and Features wizard.

2. Select Server Roles

Enable these:

- **Active Directory Domain Services**
- **DNS Server**
- **DHCP Server**

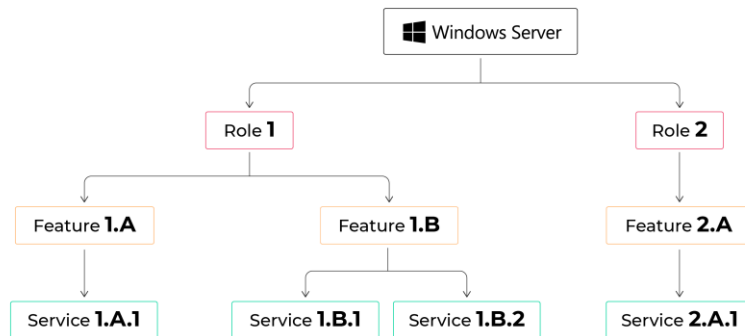


Figure 33. Selecting AD DS, DNS, and DHCP roles.

3. Complete Installation & Restart

After installation:

- Promote AD DS server to a **Domain Controller**
- Create a domain name:
- **yourlastname.local**
- Restart the server if prompted

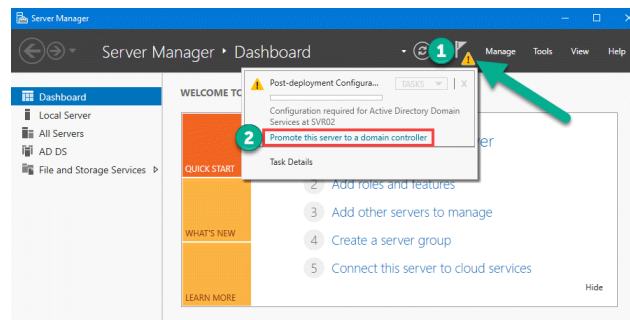


Figure 34. Promoting server to Domain Controller.



GUIDED QUESTIONS

1. Why is a static IP required before installing server roles?
2. What is the importance of Active Directory in an organization?
3. How does DNS support AD DS?
4. Why must server names follow a structured naming convention?
5. What risks occur when servers are misconfigured?

SYSTEM ADMINISTRATION AND MAINTENANCE

DHCP, DNS, AND ACTIVE DIRECTORY DOMAIN SERVICES (AD DS) CONFIGURATION

INTRODUCTION

Setting up the essential network services occurs after installing Windows Server and enabling its primary functions. Active Directory Domain Services (AD DS), DNS (Domain Name System), and DHCP (Dynamic Host Configuration Protocol) work closely together to support identity management, name resolution, and automated IP address assignment within the company.

These services facilitate the management of user logins, automate network configurations, and guarantee uninterrupted device connectivity and communication. When correctly set up, they contribute to the creation of a scalable, secure, and reliable IT infrastructure that can expand with the company.

SCOPE

This laboratory activity covers:

- Configuring DNS zones and host records
- Installing and configuring DHCP scopes
- Promoting a server to a Domain Controller (if not done previously)
- Creating Organizational Units (OUs)
- Creating domain user accounts
- Testing connectivity and domain join

MATERIALS

- Configured Windows Server VM
- Active Directory Domain Services installed
- DNS Server installed
- DHCP Server installed
- Network connection (virtual)
- Windows 10 Client VM (optional for testing)



LEARNING OUTCOMES

At the end of this activity, students should be able to:

1. Set up DNS forward lookup zones and create A records.
2. Create a DHCP scope and activate it for network use.
3. Promote the server to a Domain Controller (if it hasn't been done yet).
4. Create Organizational Units and add new domain users.
5. Connect a client computer to the domain and verify that everything is working properly.

LABORATORY ACTIVITY 4: DHCP, DNS, AND ACTIVE DIRECTORY DOMAIN SERVICES (AD DS) CONFIGURATION

Objectives:

At the end of this activity, the students should be able to:

1. Set up DNS forward lookup zones and create A records to ensure that devices within the domain can identify and communicate with one another.
2. Create and activate a DHCP scope to enable client devices to receive the necessary network settings automatically.
3. Promote the server to a Domain Controller (if it hasn't already been done) and confirm that all domain services are functioning correctly.
4. Create Organizational Units (OUs) and add new domain user accounts to help organize and manage domain resources more efficiently.
5. Join a Windows 10 client computer to the domain and check that everything works as expected, including DNS, DHCP, and Active Directory services.

BASIC INFORMATION

Here are the essential concepts students must understand before configuring DHCP, DNS, and AD DS.

What is DNS?

DNS translates domain names (e.g., server.local) into IP addresses, enabling communication between systems in a network.

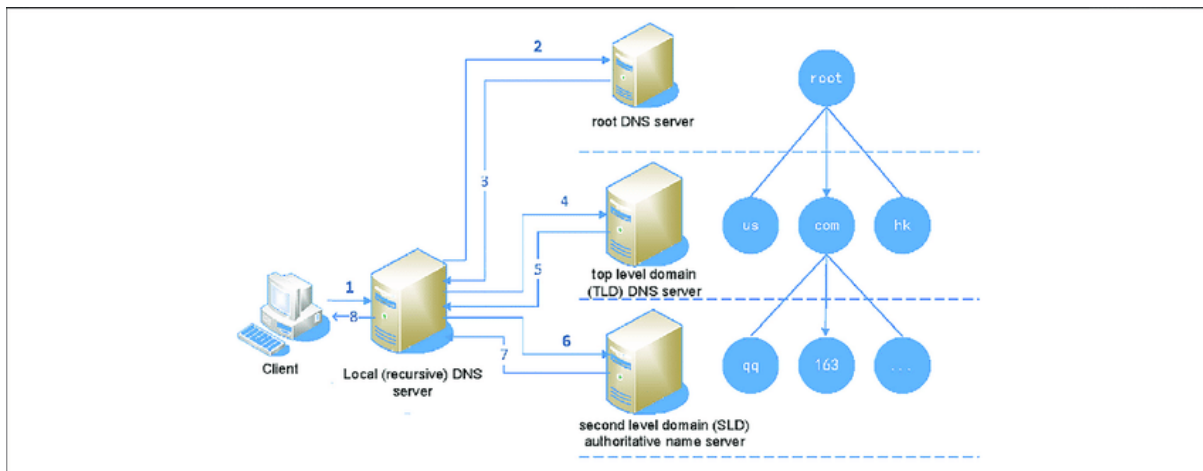


Figure 35. DNS name resolution flow.

What is DHCP?

DHCP automatically assigns IP addresses and network configurations to devices, reducing the need for manual setup.

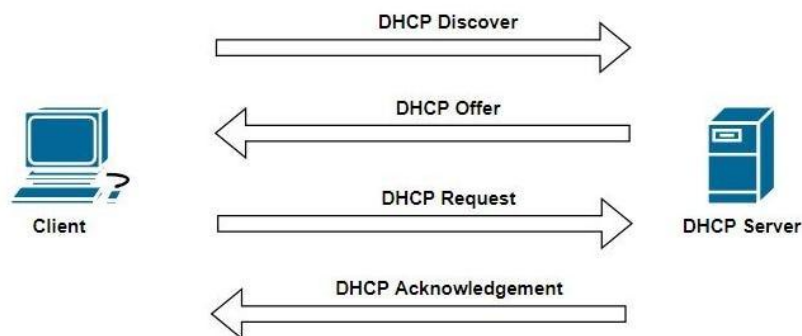


Figure 36. DHCP lease process (Discover–Offer–Request–Acknowledge).

What is Active Directory?

AD DS provides centralized user authentication, group policies, and identity management.

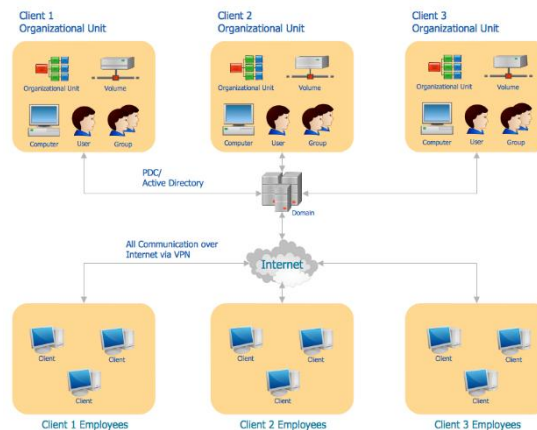


Figure 37. AD DS structure with domain, OUs, and objects.

Why Do These Three Work Together?

- **DNS** supports AD queries and domain controller location
- **DHCP** auto-assigns client IPs and optionally registers them in DNS
- **AD DS** provides centralized identity and domain control

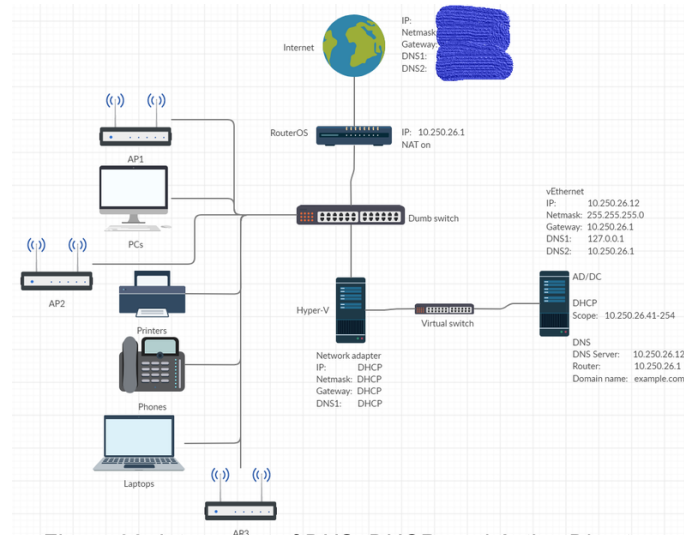


Figure 38. Integration of DNS, DHCP, and Active Directory.

TASK #1: DNS CONFIGURATION

Instructions:

1. Open DNS Manager

Launch **Server Manager** → Tools → **DNS**

2. Create Forward Lookup Zone

- Expand the server name
- Right-click **Forward Lookup Zones** → **New Zone**
- Choose **Primary Zone**
- Enter domain name:
- **yourlastname.local**
- Allow dynamic updates: **Secure only**

3. Add Host (A) Records

- Right-click inside the zone → **New Host (A)**
- Enter hostname: **server1**
- IP address: server static IP
- Add record

TASK #2: DHCP CONFIGURATION

Instructions:

1. Open DHCP Manager

Server Manager → Tools → **DHCP**

2. Create a New DHCP Scope

- Right-click **IPv4** → **New Scope**
- Scope name:
MainLAN-Scope
- IP range:
Start: 192.168.X.50
End: 192.168.X.200
- Subnet mask: 255.255.255.0
- Add router (gateway): 192.168.X.1
- Add DNS server: server IP
- Finish setup

3. Activate Scope

Right-click scope → **Activate**

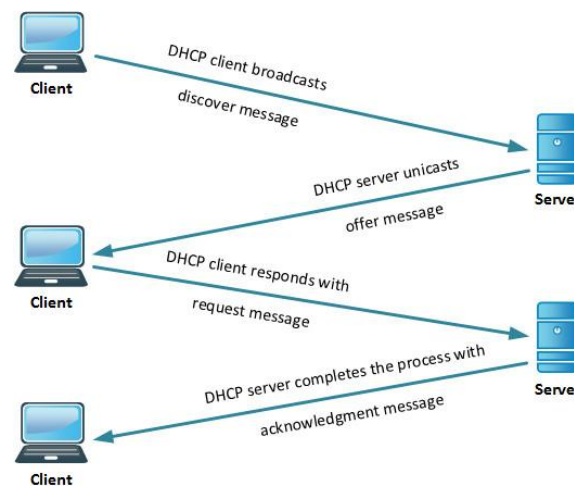


Figure 39. Creating and activating a DHCP scope.

TASK #3: ACTIVE DIRECTORY DOMAIN SERVICES CONFIGURATION

Instructions:

1. Promote Server to Domain Controller

- In Server Manager, click the flag notification at the top.
- Select "Promote this server to a domain controller."
- Choose "Create a new forest" and enter: **yourlastname.local**

- Set the DSRM password.
- Complete the installation and restart the server.

2. Create Organizational Units (OUs)

- Tools → **Active Directory Users and Computers**
- Right-click domain → **New** → **Organizational Unit**
- Create OUs:
 - Students
 - Faculty
 - Servers
 - Workstations

3. Create Domain User Accounts

- Inside **Students** OU
- Right-click → **New** → **User**
- Create accounts such as:
 - **student01**
 - **student02**

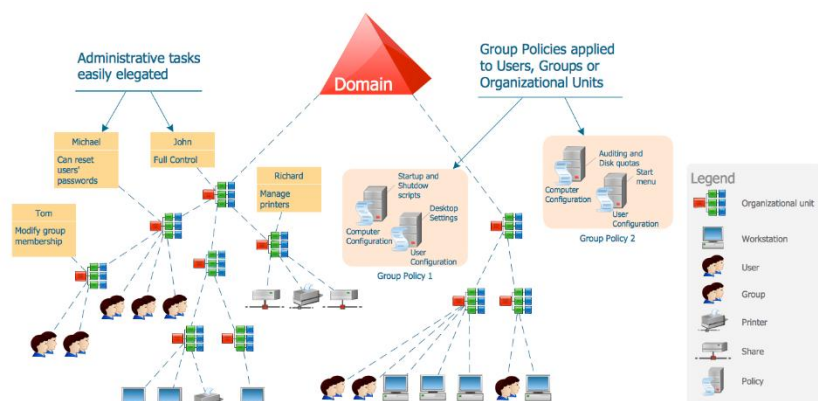


Figure 40. OU and user creation inside ADDS.

TASK #4: TESTING DHCP, DNS, AND AD DS INTEGRATION

Instructions:

Using a Windows 10 Client VM

1. Start your Windows 10 VM
2. Reset network adapter or run:

```
ipconfig /release
```

```
ipconfig /renew
```

→ Verify that DHCP assigns an IP

3. Join client to domain:

This PC → Properties → Rename this PC

4. Enter domain:

yourlastname.local

5. Log in using:

student01@yourlastname.local

If successful, the integration is working.

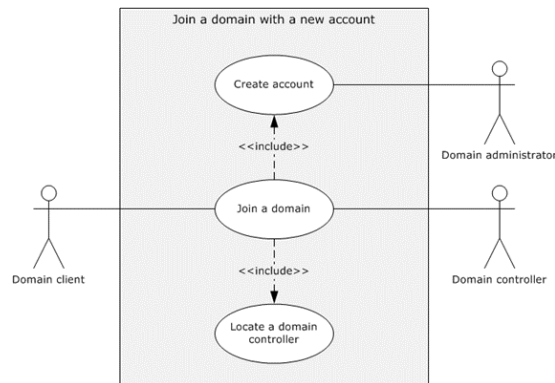


Figure 41. Windows 10 joining the domain.

GUIDED QUESTIONS

1. Why do you need to set up DNS before turning a server into a Domain Controller?
2. What problems can occur if DHCP is giving out IP addresses, but DNS isn't set up correctly?
3. Why is it important for Active Directory to have a well-organized OU structure?
4. How does joining a device to the domain make the network more secure?
5. Why are automated services like DHCP, DNS, and Active Directory so important in large organizations?

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<https://learn.microsoft.com/en-us/windows-server/identity/ad-ds/>

SYSTEM ADMINISTRATION AND MAINTENANCE

GROUP POLICY MANAGEMENT (GPO CREATION)

Q INTRODUCTION

One of the most potent features in Windows Server and Active Directory is Group Policy. System administrators can use it to manage user environments, install applications, implement security settings, and maintain consistent machine configurations throughout the domain.

Using the Group Policy Management Console (GPMC), administrators can create Group Policy Objects (GPOs) and link them to specific Organizational Units (OUs), ensuring that the correct settings are automatically applied to the appropriate users and devices.

In this activity, students will learn how to create, apply, and test Group Policy Objects in an Active Directory domain.

Q SCOPE

- Understanding Group Policy and its role in enterprise environments
- Creating and editing Group Policy Objects (GPOs)
- Linking GPOs to OUs
- Enforcing basic security policies
- Testing GPO results on Windows 10 clients

Q MATERIALS

- Windows Server Domain Controller (with AD DS installed)
- Group Policy Management Console (GPMC)
- Windows 10 domain-joined client VM
- Administrator credentials

Q LEARNING OUTCOMES

At the end of this activity, the students should be able to:

1. Navigate and use the Group Policy Management Console.
2. Create and edit a Group Policy Object.
3. Link GPOs to specific OUs.
4. Apply security and desktop configuration settings through GPO.
5. Test and verify GPO application on a Windows 10 client.

LABORATORY ACTIVITY 5: GROUP POLICY MANAGEMENT (GPO CREATION)

Objectives:

At the end of this activity, the students should be able to:

1. Access domain-wide policy settings, navigate and utilize the Group Policy Management Console (GPMC).
2. Apply configuration settings to all domain users and machines, create and modify a Group Policy Object (GPO).
3. Efficiently target policies, connect GPOs to certain Organizational Units (OUs).
4. Use Group Policy to apply desktop, security, and administrative configuration options.
5. Ensure that policies are applied correctly by testing and verifying the GPO program on a Windows 10 client.

BASIC INFORMATION

Group Policy is an essential feature in Windows Server that lets administrators control and automate settings for users and computers across a domain. With it, security rules, desktop setups, software restrictions, and network requirements can be applied automatically—without having to configure each device one by one. This helps keep large IT environments consistent, secure, and much easier to manage.

What is Group Policy?

Group Policy is a feature of Windows Server that enables administrators to centrally manage settings, security configurations, and user and computer behaviors within an Active Directory domain.

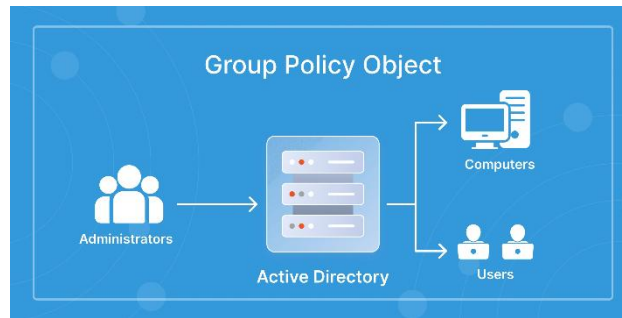


Figure 42. Overview of how Group Policy applies settings through Active Directory.

What is a Group Policy Object (GPO)?

A GPO is a set of rules and configurations that control the behavior of user accounts or computers. GPOs can enforce password policies, control desktop settings, restrict applications, and more.

Where Are GPOs Applied?

Group Policy Objects can be linked to:

- Site
- Domain
- Organizational Unit (OU)

OUs are the most common and recommended target for applying GPOs.

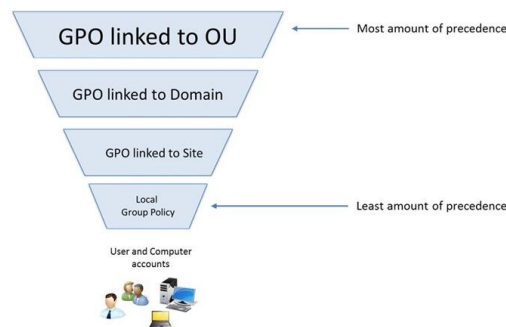


Figure 43. Order of GPO processing from Site → Domain → OU.

Why Is Group Policy Important?

Group Policy ensures consistency, security, and efficiency across the organization by automating configuration tasks.

Examples:

- Enforcing strong passwords
- Deploying software
- Restricting access to Control Panel

- Disabling USB ports
- Enforcing desktop wallpaper
- Blocking unwanted applications

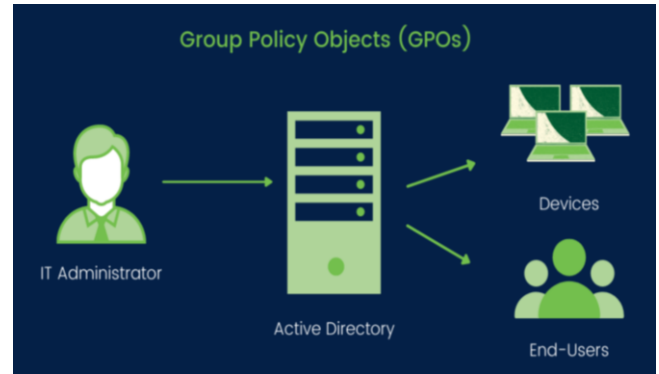
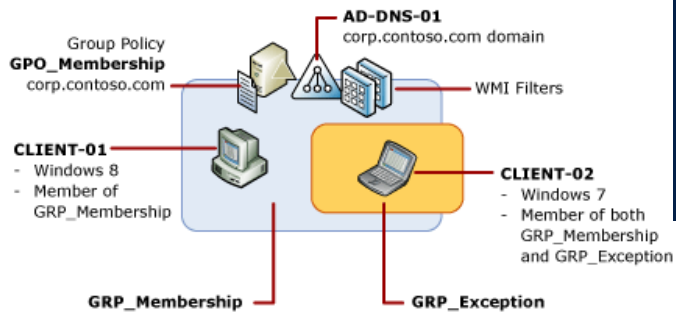


Figure 44. How Group Policy enforces settings across computers.

TASK #1: ACCESSING THE GROUP POLICY MANAGEMENT CONSOLE

Instructions:

Open GPMC

- Log in to the Windows Server Domain Controller
- Go to **Server Manager** → **Tools** → **Group Policy Management**

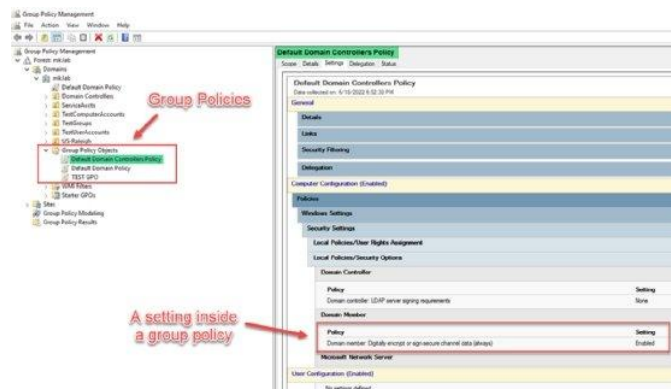


Figure 45. Opening the Group Policy Management Console (GPMC).

TASK #2: CREATING A NEW GPO

Instructions:

Create a GPO to Disable Control Panel

1. Create the GPO

- In GPMC, expand:
Forest → **Domains** → **yourlastname.local** → **Group Policy Objects**
- Right-click **Group Policy Objects** → **New**
- Name it:

Disable-ControlPanel-GPO

2. Edit the GPO

- Right-click the new GPO → **Edit**
- Navigate to:
User Configuration → Administrative Templates → Control Panel
- Double-click: **Prohibit access to Control Panel and PC Settings**
- Set to: **Enabled**
- Click **Apply** → **OK**

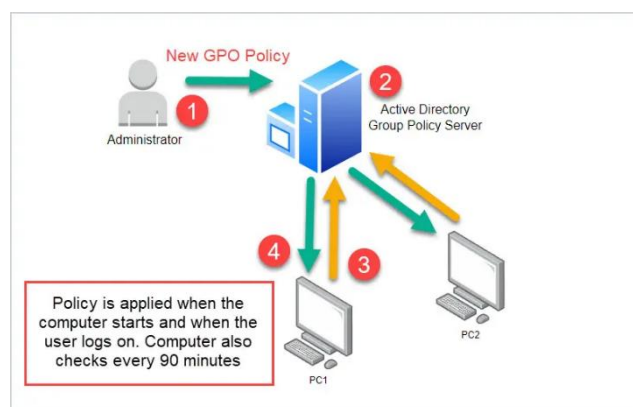


Figure 46. Creating and Editing Group Policy to disable Control Panel.

TASK #3: LINKING THE GPO TO AN ORGANIZATIONAL UNIT (OU)

Instructions:

1. Choose OU

Select an OU with test users, e.g.:

Students OU

2. Link GPO

- Right-click **Students** → **Link an Existing GPO**
- Select: **Disable-ControlPanel-GPO**

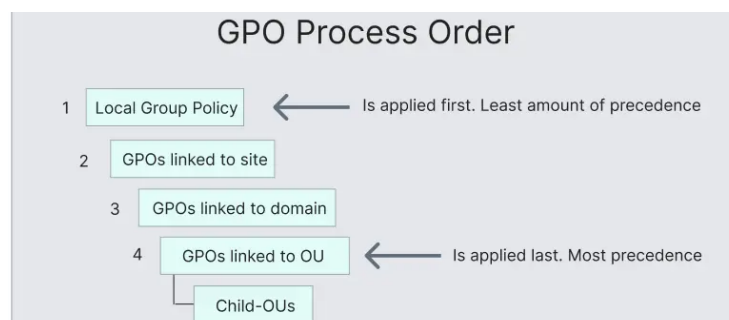


Figure 47. Linking the GPO to a specific OU.

TASK #4: REFRESHING GROUP POLICY AND TESTING ON WINDOWS 10

1. On the Windows 10 Client VM

- Log in using a domain account from the **Students** OU
- Open **Command Prompt**
- Run:

```
gpupdate /force
```

2. Test the Policy

- Attempt to open Control Panel
- Expected result: Blocked with a policy message

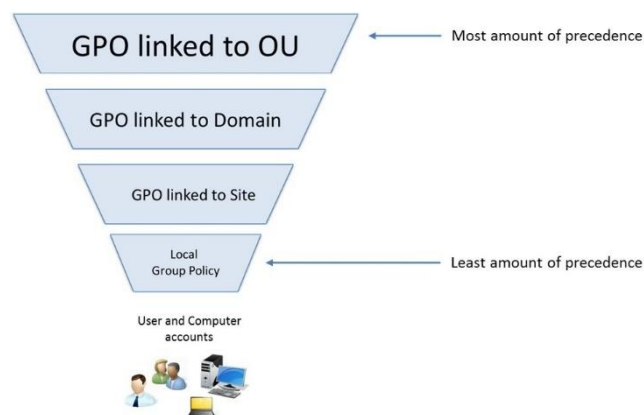


Figure 48. Group Policy applied on the client machine.

TASK #5: BONUS – CREATE A CUSTOM GPO FOR DESKTOP WALLPAPER

Instructions:

1. Create GPO: Set-Wallpaper-GPO
2. Edit:

**User Configuration → Administrative Templates → Desktop →
Desktop Wallpaper**

3. Set wallpaper path (use a shared folder)
4. Link GPO to **Students** OU
5. Test on Windows 10 VM



Figure 49. Example of applying a wallpaper via GPO.



GUIDED QUESTIONS

1. Why is it better to link GPOs to specific Organizational Units (OUs) instead of applying them at the domain level?
2. What security issues are prevented when you restrict access to the Control Panel?
3. How does Group Policy help make administrative tasks easier and more efficient?
4. What happens when two GPOs with conflicting settings apply to the same user?
5. Why is it essential to test GPOs on client computers before rolling them out to everyone?



REFERENCES

Microsoft. (2023). Group Policy overview. Microsoft Learn.
<https://learn.microsoft.com/en-us/windows-server/administration/windows-commands/group-policy>

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Rountree, D., & Castrillo, I. (2013). The basics of IT auditing. Syngress.

SYSTEM ADMINISTRATION AND MAINTENANCE

FILE SHARING, NTFS PERMISSIONS, AND NETWORK SHARES

Q INTRODUCTION

Organizations rely on file sharing to keep their files in one place, manage who can access specific data, and make collaboration easier. With Windows Server, administrators can set up shared folders, configure NTFS and share permissions, and give the right domain users and groups the access they need.

In this activity, students will learn how to create shared resources, apply the correct permissions, and check access from client computers to make sure everything is functioning as expected.

Q SCOPE

- Creating file server folder structures
- Applying NTFS permissions
- Setting up Windows network shares
- Mapping shared drives
- Testing access using domain users

Q MATERIALS

- Windows Server VM (with AD DS)
- Windows 10 domain-joined client VM
- Domain user accounts (Faculty, Students, Admin)
- Administrator credentials
- Word processor for documentation

Q LEARNING OUTCOMES

At the end of this activity, the students should be able to:

1. Create organizational file-sharing structures.
2. Configure appropriate NTFS permissions.



3. Create and manage network shares.
4. Access shares from a Windows client and map network drives.
5. Analyze and verify effective permissions.

LABORATORY ACTIVITY 6: FILE SHARING, NTFS PERMISSIONS, AND NETWORK SHARES

Objectives

At the end of this activity, the students should be able to:

1. Enable effective access and administration, provide a well-organized folder structure for departmental or user-level file sharing.
2. Depending on user responsibilities and access needs, set up the proper NTFS permissions.
3. Use share permissions and follow correct naming standards to create and manage Windows Server network shares.
4. Can successfully map network drives using UNC paths and access shared folders from a Windows client.
5. Ensure that the NTFS and share permissions give each user or group the appropriate amount of access, analyze and confirm effective permissions.

BASIC INFORMATION

File sharing is one of the core services provided by Windows Server. Organizations rely on shared folders to store documents, collaborate on projects, and control access to sensitive information. Windows Server provides two key layers of access control—**NTFS permissions** and **share permissions**—which together determine the effective access granted to users and groups. Understanding how these permissions interact is essential for secure and efficient file management in a domain environment.

File Sharing in Windows Server

Windows Server allows the creation of centralized folders that users can access over the network using SMB (Server Message Block) protocol.

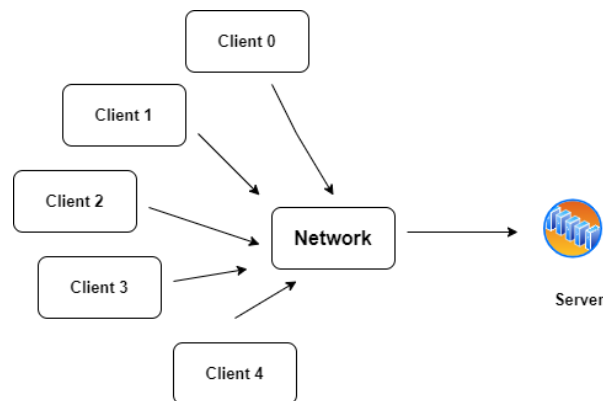


Figure 50. Basic file sharing architecture in a Windows Server environment.

Understanding NTFS Permissions

NTFS (New Technology File System) permissions control access to files and folders at the operating system level, regardless of how the user accesses them.

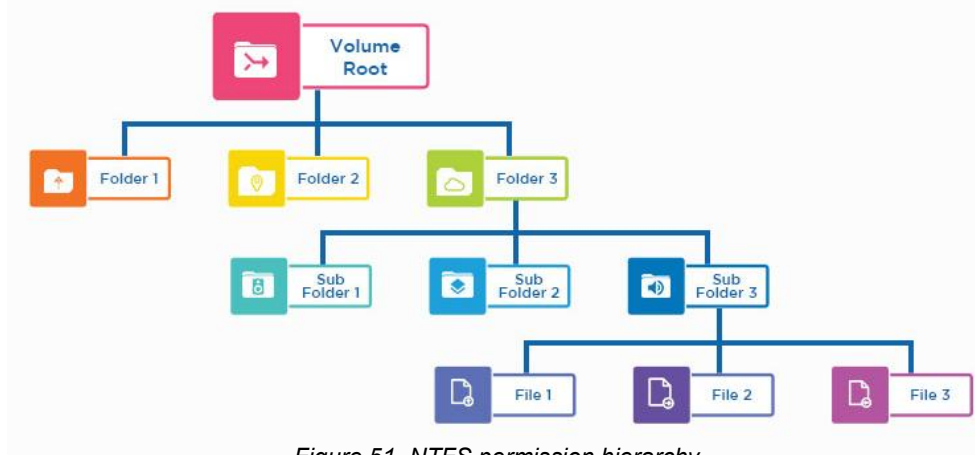


Figure 51. NTFS permission hierarchy.

Share Permissions vs. NTFS Permissions

Share permissions apply only when folders are accessed over the network. When both apply, **effective permission = most restrictive between NTFS and Share permissions**.

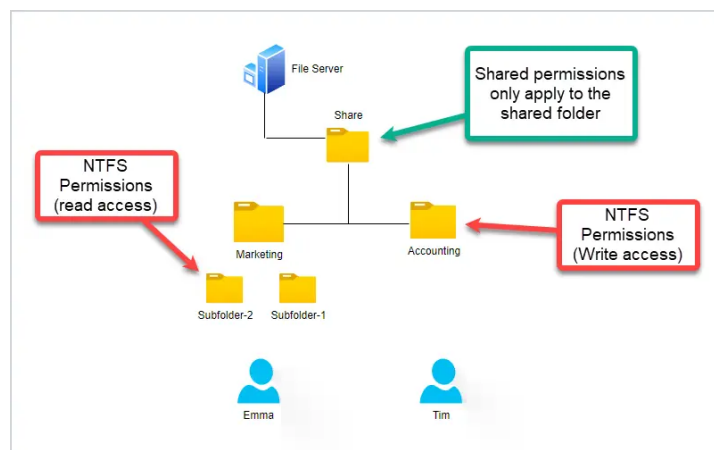


Figure 52. Relationship between NTFS and share permissions.

UNC Paths and Network Drive Mapping

UNC (Universal Naming Convention) paths allow access to shared folders using the format:

`\\ServerName\ShareName`

Network drives map folders to drive letters for easy access.

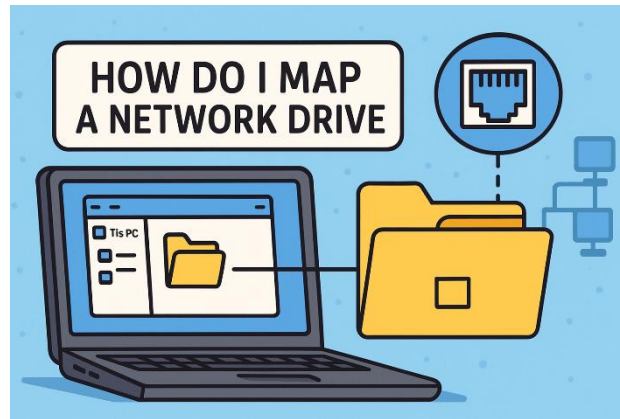


Figure 53. UNC path and network drive mapping.

TASK #1: CREATE FOLDER STRUCTURE FOR SHARED DATA

Instructions:

1. Log in to your Windows Server as Administrator.
2. Open File Explorer → Create folder:
C:\Shares
3. Inside it, create:
 - Public
 - Dept_Faculty
 - Dept_Students
 - Private_Admin
4. Take a screenshot of the structure.

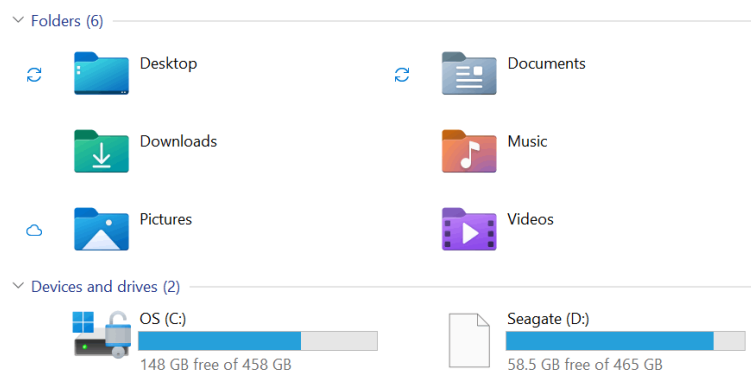


Figure 54. Example folder structure for shared resources..

TASK #2: CONFIGURE NTFS PERMISSIONS

Instructions:

Assign permissions based on organizational roles.

A. Public Folder

- Everyone → **Read & Execute**
- Administrators → **Full Control**

B. Faculty Folder

- Faculty_Group → **Modify**
- Administrators → **Full Control**

C. Students Folder

- Students_Group → **Modify**
- Administrators → **Full Control**

D. Private_Admin Folder

- Administrators → **Full Control**
- Remove unnecessary entries

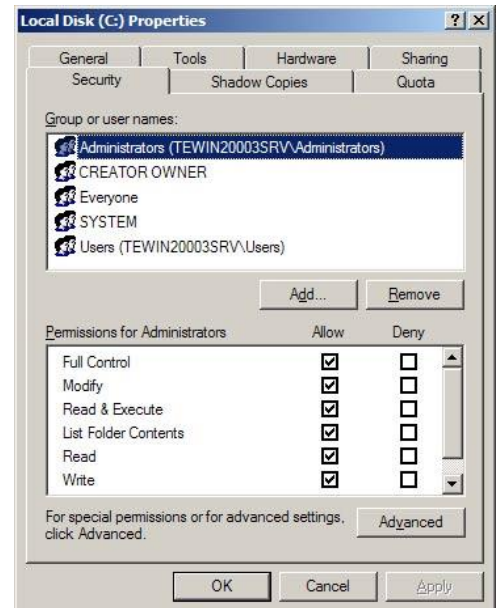


Figure 55. NTFS permissions configuration window..

TASK #3: CREATE NETWORK SHARES

Instructions:

1. Right-click each folder and open **Properties** → **Sharing** → **Advanced Sharing**.
2. Set the share names: Public, Faculty, Students, and AdminPrivate.
3. Apply the following share permissions:
 - **Public:** Everyone – Read
 - **Faculty:** Faculty_Group – Change, Read
 - **Students:** Students_Group – Change, Read
 - **AdminPrivate:** Admin – Full Control
4. Check your work in **Computer Management** → **Shared Folders** → **Shares**.

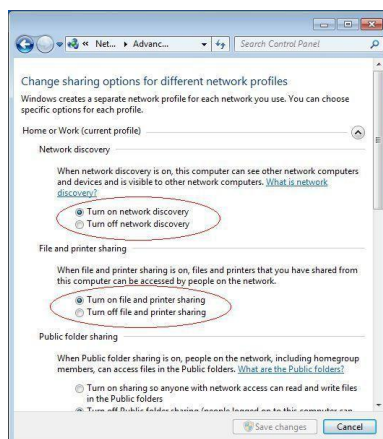


Figure 56. Advanced Sharing and Shared Folders interface.

TASK #4: ACCESS SHARES FROM A CLIENT MACHINE

Instructions (Windows 10 Client):

1. Sign in using a Student account.
2. Open the Run box and enter:
`\SRV-LastName`
3. Check each shared folder:
 - Can you open **Public**?
 - Can you add or edit files in **Students**?
 - Is access to **Faculty** blocked?
 - Is **AdminPrivate** denied?
4. Sign out and repeat the same checks using a **Faculty account**.
5. To map a network drive, right-click **This PC** and choose **Map Network Drive**.
 - Example:

Drive: H: → \SRV-LastName\Students

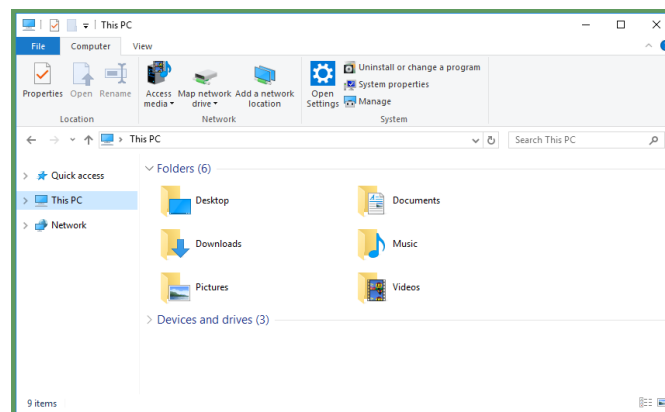


Figure 57. Accessing shared folders and mapped drives.

TASK #5: EFFECTIVE PERMISSIONS ANALYSIS

Instructions:

Complete the following table using your test results:

User	Share	Expected Access	Test Result	Notes
Student01	Public	Read		
Student01	Students	Modify		
Student01	Faculty	No Access		
Faculty01	Faculty	Modify		
Faculty01	Students	No Access		
Any non-admin	AdminPrivate	No Access		

Then write a **150–200 word explanation** of:

- How NTFS + Share permissions combine
- Why do effective permissions use the **most restrictive**
- Importance of proper permission planning.

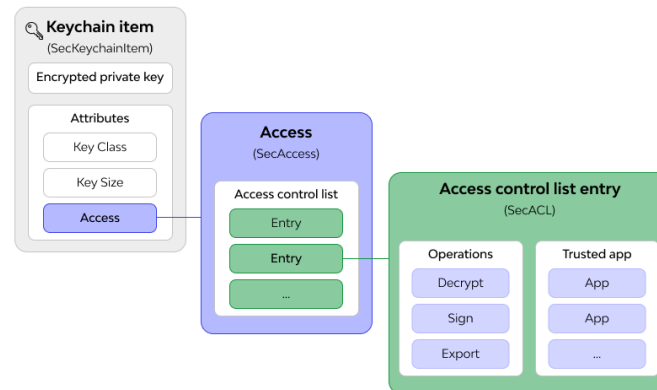


Figure 58. Conceptual diagram of effective permissions



GUIDED QUESTIONS

1. Why is it better to assign permissions to groups instead of individual users?
2. How do NTFS permissions differ from share permissions?
3. What could cause a user to have less access than they think they should?
4. What problems can happen if permissions are set incorrectly?
5. Why is the “least privilege” approach important when setting up a file server?



REFERENCES

Microsoft. (2023). NTFS permissions overview. Microsoft Learn.
<https://learn.microsoft.com/en-us/windows/security/identity-protection/access-control/ntfs-permissions>

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Rountree, D., & Castrillo, I. (2013). The basics of IT auditing. Syngress.



SYSTEM ADMINISTRATION AND MAINTENANCE

WINDOWS FIREWALL & SECURITY HARDENING

INTRODUCTION

Security hardening entails configuring the server to decrease vulnerabilities and safeguard it against potential attacks. Windows Firewall, User Account Control (UAC), and account policies all contribute to the security of the operating system and its services. In this assignment, students will learn how to set up firewall rules, implement security policies, and employ basic hardening techniques to increase server security.

SCOPE

- Windows Firewall with Advanced Security
- Creating inbound and outbound firewall rules
- Enabling/Disabling features securely
- Applying password policies
- Local and domain-level security baselines.

MATERIALS

- Windows Server VM
- AD DS installed (optional but recommended)
- Administrator credentials
- Windows 10 client for testing

LEARNING OUTCOMES

At the end of this activity, the students should be able to:

1. Navigate Windows Firewall with Advanced Security.
2. Create and apply custom firewall rules.
3. Implement password and account lockout policies.
4. Harden server settings using best practices.
5. Test and validate applied security configurations.

LABORATORY ACTIVITY 7: WINDOWS FIREWALL & SECURITY HARDENING

Objectives:

At the end of this activity, the students should be able to:

1. Explore Windows Firewall with Advanced Security and become familiar with its main sections and rule types.
2. Create and apply your own inbound and outbound rules to control network traffic and secure server communication.
3. Set up password and account lockout policies to strengthen login security.
4. Improve the server's security by disabling services that aren't needed and applying recommended security settings.
5. Test the changes you made to confirm that the firewall rules and security policies are working correctly.

BASIC INFORMATION

Windows Firewall and system hardening are critical components of protecting a Windows Server environment. Servers are continually vulnerable to various threats, including unauthorized access, malware, network attacks, and misconfiguration. System administrators can utilize firewall rules, security policies, and hardening measures to mitigate vulnerabilities, manage access, and maintain a stable and secure server environment.

Windows Firewall Overview

Windows Firewall controls inbound and outbound traffic, preventing unauthorized access and limiting attack exposure.

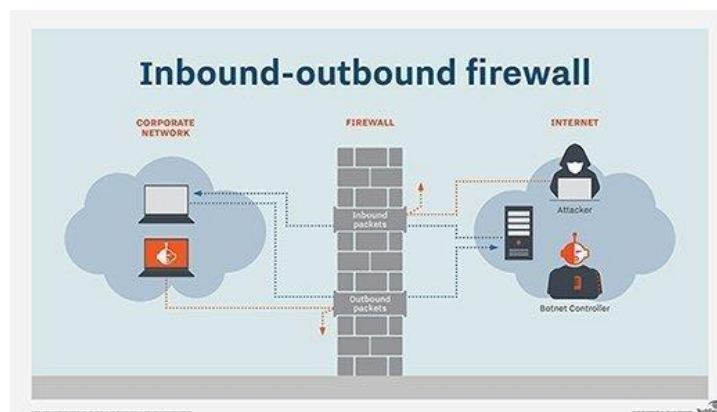


Figure 59. Windows Firewall architecture.

Inbound vs. Outbound Rules

- **Inbound:** Controls incoming traffic to the server.
- **Outbound:** Controls outgoing traffic from the server.

Security Hardening Concepts

Security hardening includes:

- Turning off unused services
- Enforcing password policies
- Limiting administrative privileges
- Enabling secure protocols



Figure 60. Example steps in system hardening.

TASK #1: EXPLORING WINDOWS FIREWALL WITH ADVANCED SECURITY

Instructions:

1. Open **Server Manager** → **Tools** → **Windows Firewall with Advanced Security**
2. Identify:
 - Inbound Rules
 - Outbound Rules
 - Connection Security Rules
 - Monitoring
3. Take a screenshot.

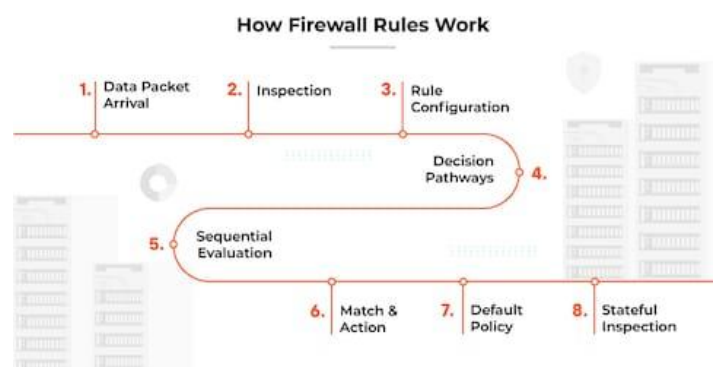


Figure 61. Windows Firewall with Advanced Security console.

TASK #2: CREATING CUSTOM INBOUND FIREWALL RULES

Example: Allow RDP only for administrators

1. Go to **Inbound Rules** → **New Rule**
2. Select **Port** → Next
3. Protocol: **TCP**, Port: **3389**
4. Allow the connection
5. Apply to Domain / Private
6. Name: **RDP-Admins-Only**

Verify rule appears and is active.

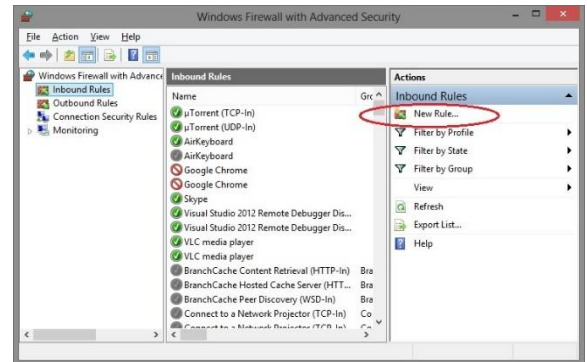


Figure 62. Creating an inbound rule.

TASK #3: IMPLEMENTING PASSWORD POLICIES

Instructions:

1. Open **Group Policy Management**
2. Edit:
Default Domain Policy → **Computer Configuration** → **Policies** →
Windows Settings → **Security Settings** → **Account Policies**
3. Configure:
 - Minimum Password Length: 8–12
 - Password Complexity: Enabled
 - Maximum Password Age: 30 days
4. Apply GPO.

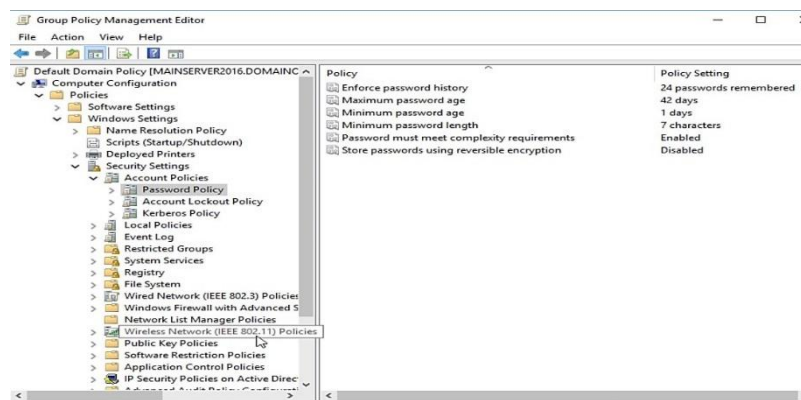


Figure 63. Example password policy settings.

TASK #4: IMPLEMENT ACCOUNT LOCKOUT POLICY

Instructions:

1. Open the same GPO

2. Configure:

- Account Lockout Threshold: 3 attempts
- Lockout Duration: 10–15 minutes

3. Apply and run `gpupdate /force` on client.

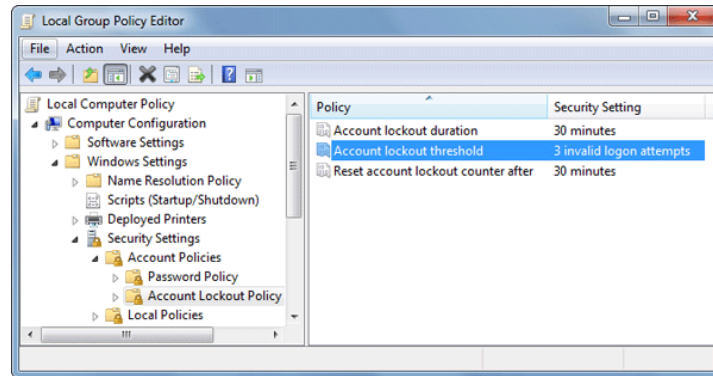


Figure 64. Account lockout configuration.



GUIDED QUESTIONS

1. Why should servers have strict firewall rules?
2. Why is password complexity important?
3. What risks occur if account lockout is not configured?
4. How does hardening reduce attack surfaces?
5. Why should administrators NOT use their admin accounts for daily tasks?



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SYSTEM ADMINISTRATION AND MAINTENANCE

WINDOWS BACKUP & RESTORE

INTRODUCTION

Backup and restore capabilities safeguard data and system integrity in the event of corruption, human error, hardware failure, or cyberattacks. Windows Server provides built-in backup tools that allow system state backup, file-level backup, scheduled backups, and restore operations.

SCOPE

- Installing Windows Server Backup
- Creating full system backups
- Performing file backups
- Creating scheduled backups
- Performing restore operations.

MATERIALS

- Windows Server VM
- Secondary virtual hard disk (recommended)
- Administrator credentials

LEARNING OUTCOMES

At the end of this activity, the students should be able to:

1. Install and launch Windows Server Backup.
2. Perform full and custom backups.
3. Configure scheduled backups.
4. Restore files and system states.

LABORATORY ACTIVITY 8: WINDOWS BACKUP & RESTORE

Objectives:

At the end of this activity, the students should be able to:

1. Install and start Windows Server Backup as a server function.
2. Create full and customized backups, including file-level and system-state backups.
3. Configure scheduled backups to automate regular data protection and maintain data security.
4. Use the various backup sets and recovery tools to restore data and system states.

BASIC INFORMATION

Windows Server Backup plays a practical role in protecting system data and keeping business operations running when problems arise, such as data loss, system errors, or hardware failure. With proper backup and recovery processes, an organization can restore important files, bring systems back to working condition, and reduce downtime during unexpected disruptions. For system administrators, being able to configure, run, schedule, and restore backups is a necessary part of managing and maintaining a reliable server environment.

What is Windows Server Backup?

A built-in tool that provides backup and recovery for Windows Server environments.

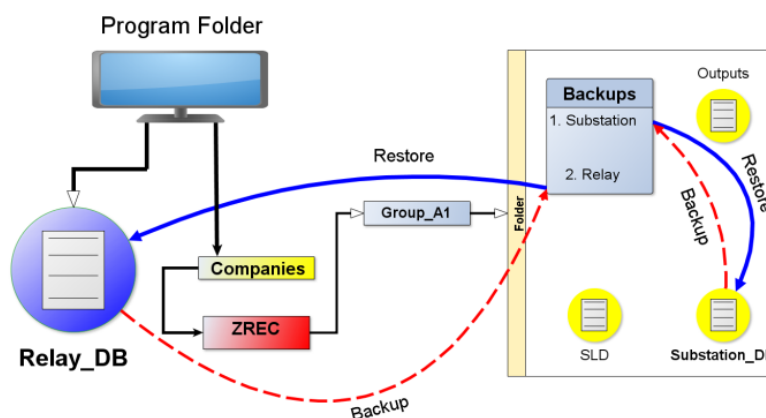


Figure 65. Windows Server Backup interface.

Types of Backups

- Full Backups
- Custom Backups
- System State Backups

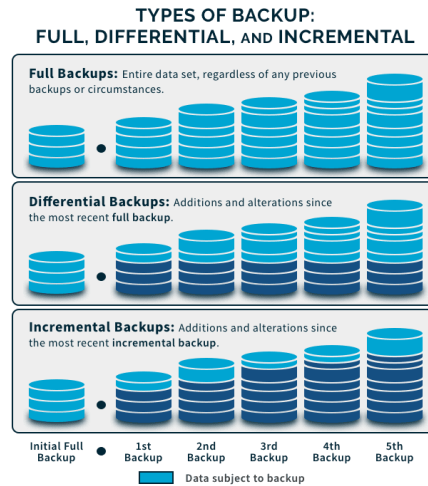


Figure 66. Overview of Backup types.

Restore Operations

Restore from:

- Backup disk
- Network share
- System state backup

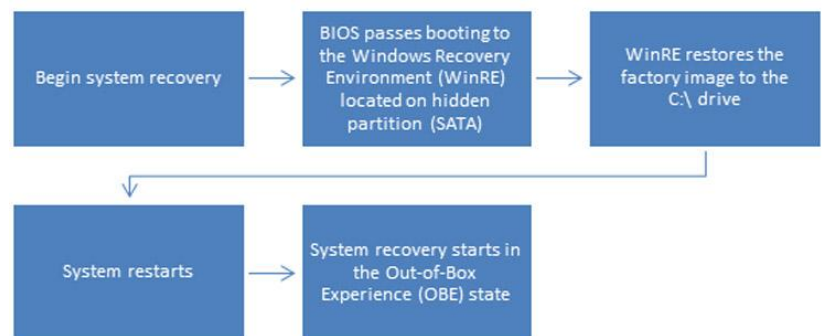


Figure 67. Restore workflow.

TASK #1: INSTALL WINDOWS SERVER BACKUP

Instructions:

1. Open **Server Manager**
→ **Add Roles and Features**
2. Install **Windows Server Backup** feature
3. Launch tool via **Tools**
→ **Windows Server Backup**

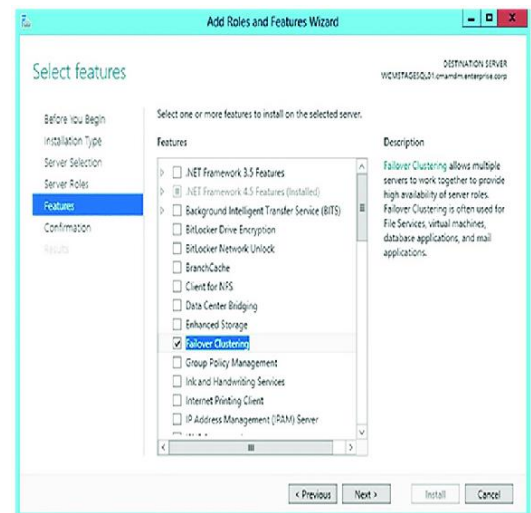


Figure 68. Installing Windows Server Backup

TASK #2: PERFORM A FULL SERVER BACKUP

Instructions:

1. Open Windows Server Backup
2. Click **Backup Once**

3. Choose **Full Server**
4. Save to:
 - Another virtual disk, or
 - Network share
5. Start backup and take a screenshot.

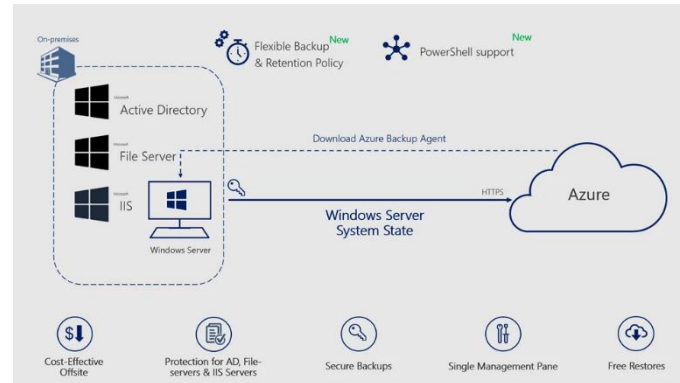


Figure 69. Full server backup process.

TASK #3: CONFIGURE SCHEDULED BACKUP

Instructions:

1. Click **Backup Schedule**
2. Choose **Full Server** or **Custom**
3. Set schedule (daily recommended)
4. Select destination
5. Confirm setup

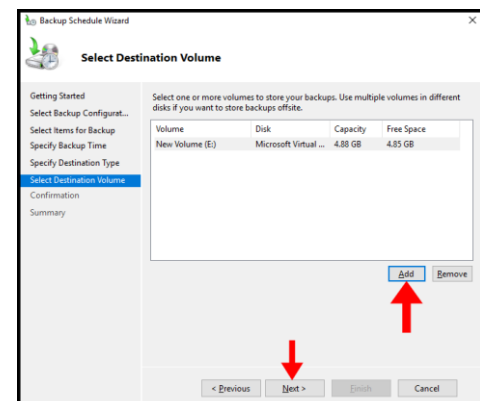


Figure 70. Scheduled backup configuration.

TASK #4: RESTORE FILES OR FOLDERS

Instructions:

1. Open Windows Server Backup
2. Click **Recover**
3. Choose a backup location
4. Select file(s) to restore
5. Restore to the alternate location
6. Document before/after results

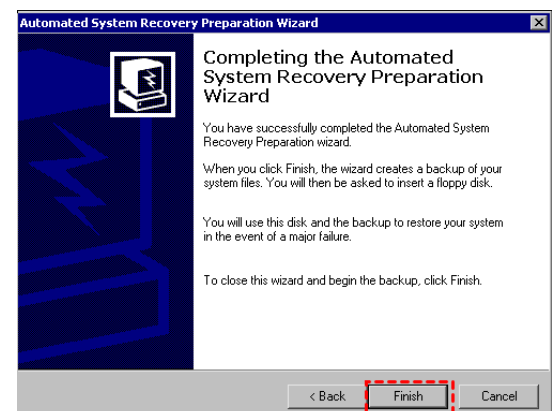


Figure 71. File restore wizard.



GUIDED QUESTIONS

1. Why should backups be stored on separate disks?
2. What is the importance of scheduled backups?
3. What is a system state backup?
4. When should a restore operation be performed?
5. What risks occur when backups are not secured?



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SYSTEM ADMINISTRATION AND MAINTENANCE

ADVANCED NTFS (AUDITING & QUOTAS)

INTRODUCTION

Advanced NTFS features such as **auditing** and **disk quotas** provide deeper control over file access and storage use. Auditing tracks user actions on folders/files, while quotas limit disk usage per user to prevent storage abuse.

SCOPE

- Enabling NTFS auditing
- Configuring audit policies via Group Policy
- Reviewing audit logs
- Enabling disk quotas on NTFS volumes
- Setting per-user storage limits

MATERIALS

- Windows Server VM
- Windows 10 domain-joined client
- Event Viewer
- NTFS volume with free space

LEARNING OUTCOMES

At the end of this activity, the students should be able to:

1. Enable NTFS auditing on files/folders.
2. Apply audit policies using GPO.
3. Analyze audit logs in Event Viewer.
4. Configure and test disk quotas.

LABORATORY ACTIVITY 9: ADVANCED NTFS (AUDITING & QUOTAS)

Objectives:

At the end of this activity, the students should be able to:

1. Turn on NTFS auditing for specific files and folders to track who accesses them and what changes are made.
2. Use Group Policy to apply audit policies so access and activity can be logged consistently across the domain.
3. Review the audit logs in Event Viewer to see access attempts, file changes, and other security-related events.
4. Set up and test disk quotas to control how much storage users can use and to monitor disk space on NTFS drives.

BASIC INFORMATION

In Windows Server settings, NTFS auditing and disk quotas allow administrators more control over file access and storage use. Disk quotas control how much space each user may use, while inspecting file access and activity for security, compliance, and troubleshooting. When used collectively, these capabilities improve data management and help maintain the file server safe and reliable.

NTFS Auditing Overview

Auditing tracks changes (create, delete, modify) made by users and logs them in the Event Viewer.

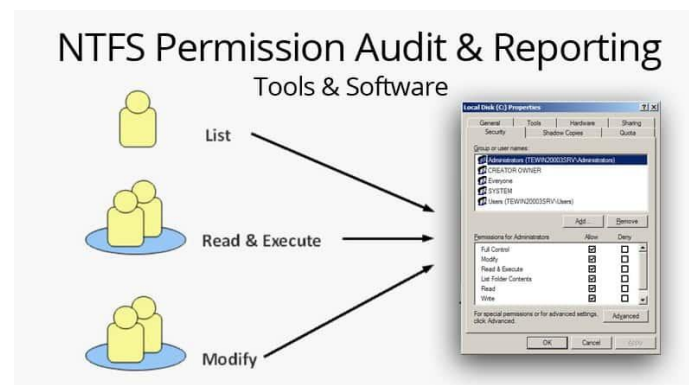


Figure 72. Windows Server Backup interface.

Disk Quotas

Disk quotas limit storage to avoid space exhaustion.

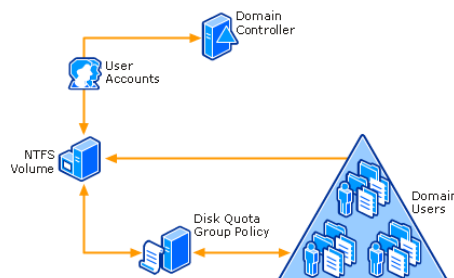


Figure 73. Disk quota management.

TASK #1: ENABLE FOLDER AUDITING

Instructions:

1. Create folder:
C:\AuditTest
2. Right-click → Properties → Security → Advanced → **Auditing**
3. Add: **Students_Group**
4. Enable:
 - Create files
 - Delete files
 - Modify

TASK #2: CONFIGURE AUDIT POLICY IN GROUP POLICY

Instructions:

1. Open **GPMC**
2. Edit:
Default Domain Controllers Policy
3. Navigate to:
Computer Configuration → Policies → Windows Settings
→ Security Settings → Advanced Audit Policy Configuration
→ Object Access
4. Enable:
 - Audit File System → Success & Failure

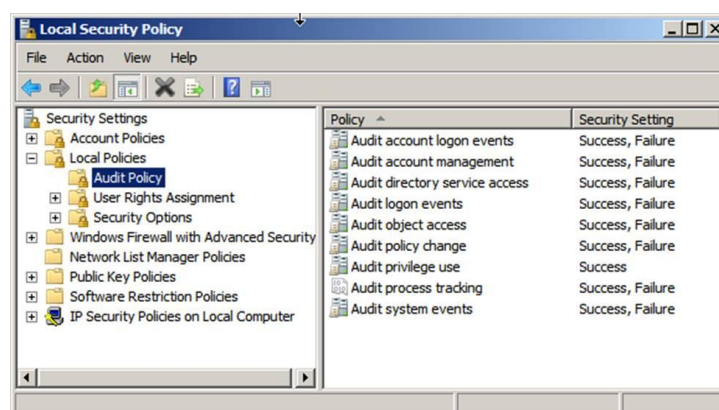


Figure 74. Enabling advanced audit policy.

TASK #3: GENERATE AND REVIEW AUDIT LOGS

Instructions:

1. Log in as a student account

2. Access \\SRV-LastName\AuditTest
3. Create and delete files
4. On server, open **Event Viewer**
5. Go to:
Windows Logs → Security
6. Filter by Event IDs: **4663, 4656**

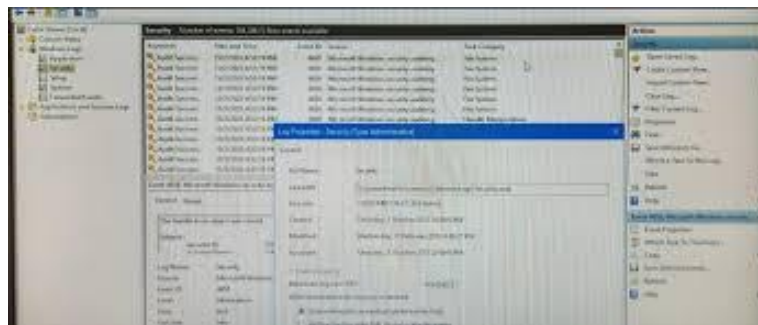


Figure 75. Reviewing audit logs.

TASK #4: CONFIGURE DISK QUOTAS

Instructions:

1. Right-click NTFS drive (C: or D:) → **Properties** → **Quota**
2. Enable Quota Management
3. Limit users to **500 MB**
4. Deny disk space to users exceeding the limit
5. Test with a student account by copying large files

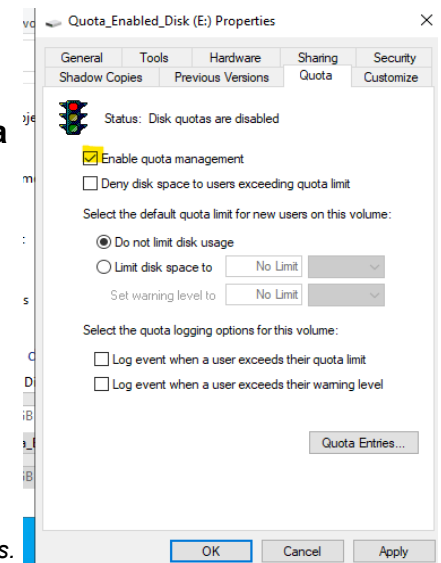


Figure 76. Configuring disk quotas.

GUIDED QUESTIONS

1. Why is auditing important in secure environments?
2. What risks arise if quotas are not implemented?
3. What are the limitations of NTFS auditing?
4. How can audit logs help in investigations?
5. Why should quota thresholds be monitored regularly?



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Engr. Raymond Bryan M. Esmenda is a faculty member of the Information Technology Department at Pangasinan State University – Lingayen Campus. He is a licensed Electronics Engineer and holds a Master's Degree in Management Engineering from the PSU School of Advanced Studies. His background in both engineering and information technology provides a strong foundation for his work in teaching system administration.

He has taught a range of courses related to IT infrastructure and security, including Basic Computer Networking, Advanced Networking, and Information Assurance and Security. His experience enables him to connect classroom concepts with real-world administrative tasks, such as server management, network configuration, system maintenance, and security hardening, making his instruction practical and relevant for laboratory learning.

Over the years, Engr. Esmenda has contributed to curriculum development, accreditation efforts, and the improvement of instructional materials that support hands-on training in system administration. He has also participated in university initiatives focused on digital transformation, IT governance, and quality assurance, helping to ensure that students acquire skills aligned with industry needs.

His professional interests include network infrastructure design, server and systems administration, information security management, and the use of technology in organizational development. Engr. Esmenda remains committed to creating a learning environment that values technical skill, ethical responsibility, and continuous improvement—principles reflected in this System Administration and Maintenance Laboratory Manual.